System overview: locations and cable bundles

Where each part lives and which cables share each run (from the model's runs, E8). Circuits on E2–E7, E9 and E16; routes to scale on E8, E10–E15.

Roof platform
PV1 190 W + PV2 130 W, PV5 visor 250 W
ST1 Starlink, U11 visor controller (E9)

internal: PVV

6 CABLES
DO4, PVR, PVS, VIS, VSW, VIG

Tent (detachable)
services box: XI1 inlet, Q4 RCBO, XS3 outlet, X5
PV3 + PV4 2 × 100 W flexible (series)

internal: T2, WHT

2 CABLES
T1A, T1D

Engine bay
BT0 start battery + F8 60 A MIDI
FBX fuse box: F9 2 A add-a-fuse (IGN)

Gap junction box
J1–J4 keyed connectors (tent)
barrier 230 V | 12 V, roof pass-through

2 CABLES
ALT, IGN

10 CABLES
AC4, AC5, DO4, DO5, PVR, PVT
PVS, VIS, VSW, VIG

PASSENGER CHASSIS RAIL

15 CABLES
AC4, AC5, PE2, DO3, DO4, DO5
ALT, IGN, PVR, PVT, PVS, CHB
VIS, VSW, VIG

2 CABLES
ALT, IGN

3 CABLES
DO3, ALT, IGN

13 CABLES
AC4, AC5, DO3, DO4, DO5, ALT
IGN, PVR, PVT, PVS, VIS, VSW
VIG

Factory-outlet gland
passenger rear tub wall

1 CABLE
DO3

18 CABLES
AC1, AC3, AC4, AC5, DCF, ALT
IGN, PVR, PVT, PVS, SW1, CHB
SW2, D1, D2, D4, D5, VIG

Cabin, under the rear seat
X4 SB50 inlet → U12 compute box
HA PC, router, PoE switch

Wall drop + tray
drop beside the gland; tray under the lid

17 CABLES
AC1, AC3, AC5, DCF, ALT, IGN
PVR, PVT, PVS, SW1, CHB, SW2
D1, D2, D4, D5, VIG

16 CABLES
AC1, AC3, AC4, DCF, PE2, DO3
DO4, DO5, SW1, SW2, D1, D2
D4, D5, VIS, VSW

TUB

Front bay
BT1/BT2, F1/F2, BB1–BB4, QS1, P1, F3
U1 MultiPlus-II, U2 Orion, U3/U4/U13 MPPTs
F4–F7 MIDI/MEGA, F17 MRBF (pod), M1 fan
TH1 + HP1/HP2 heaters, F18, foam (E16)

internal: B1P, B1N, B2P, B2N, RB1, RB2, SH1, SH2, CT1, INVP, INVN, ORO, MPR, MPT, MPV, RBF, TS1, HTR, HTP, HTS, FAN, PE1

Rear control cabinet
FB1 fuse block, RCBO enclosure (Q0–Q3)
U5 Cerbo GX, rear panel: U6, S1–S3,
XS1 GPO, X1/X2, U7/U8, X6

internal: AC2, DO1A, DO1B, DO1C, DO1D, DO1E, CGX, D3

2 CABLES
AC3, DO2

Power slide pod
FB2, XS2 GPO, X3 Anderson, X7 12 V

ROUTING RULES

- Every cable that leaves the tub goes out through the factory 240 V outlet hole (passenger rear wall) and along the passenger chassis rail.
- Nothing passes through the load lid.
- The tent harness (dashed) unplugs at the gap junction, so the tent lifts off with its services box and solar.
- All switching is at the rear panel: battery isolate (S1 → QS1), inverter on/off (S2), visor (S3), GX Touch 50.
- 230 V never shares a conduit or gland row with 12 V.
- Line weight ∝ number of cables in the run.

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E1 of E16**
TITLE **System overview: locations and cable bundles**	REV / DATE **v0.14 · 2026-10-23**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

DC power: battery bank and main distribution

Front bay, between the wheel arches. 12.8 V LiFePO$_4$ bank → isolation → protection → system busbars.

FRONT BAY
tub front wall, between the arches (lid can't open with the tent on; everything is serviced from the rear)

RBF · 1.0 mm² → S1 +12 V feed
always live (bank side) · sheet E6

F10 2 A

QS1 Battery switch
Blue Sea ML-RBS 7713
500 A latching, manual knob
IN · OUT · CTRL

SW1 · 4-core 1.0 mm² → S1 rear panel
sheet E6

BB1 Bank + busbar

B1P · 70 mm²

B2P · 70 mm²

RB1 · 120 mm²

RB2 · 120 mm²

F1 MRBF 300 A

F2 MRBF 300 A

θ sensor

+ M8

+ M8

BT1 Battery 1
iTechworld iTech400X Pro
12.8 V · 400 Ah LiFePO$_4$
stands on end, terminals rear
internal BMS (Bluetooth only)

BT2 Battery 2
iTechworld iTech400X Pro
12.8 V · 400 Ah LiFePO$_4$
stands on end, terminals rear
internal BMS (Bluetooth only)

F11

− M8

− M8

B1N · 70 mm²

B2N · 70 mm²

TS1 · temp sensor 2-core

D2 · VE.Direct → U5 Cerbo
sheet E6

V+ · AUX · VE.D

P1 Battery monitor
Victron SmartShunt 500 A
all − current passes it
Aux = battery temp (TS1)
BATT · SYS

BB2 Bank − busbar

SH1 · 120 mm²

SH2 · 120 mm²

BB3 System + busbar

F17 MRBF 30 A

DO2+ → power-slide pod
6 mm² high-flex · chain · sheet E3

F4 MIDI 70 A

F5 MIDI 40 A

F6 MIDI 40 A

F6 (shared)

ORO+ ← U2 Orion XS
16 mm² twin · sheet E4

MPR+ ← U3 MPPT 100/30
16 mm² twin · sheet E4

MPT+ ← U4 MPPT 75/15
6 mm² twin · sheet E4

MPV+ ← U13 MPPT 100/20
6 mm² twin · sheet E4

DCF+ → FB1
35 mm² twin · E3

F7 MEGA 100 A

CT1 · 2×50 mm²

F3 Class-T 400 A

INVP · 2×50 mm²

DC+

U1 Inverter / charger
Victron MultiPlus-II 12/3000/120-32
230 V · 2400 W cont. · 120 A charge
DC: 2 × 50 mm² per pole (0–5 m)
AC-in / AC-out → sheet E5
VE.Bus, remote on/off → sheet E6
lies flat beside the batteries

F12 5 A

F18 10 A

HTR+ → TH1 battery heater
2.5 mm² twin · sheet E16

M

M1
fan 92 mm

FAN · 1.5 mm² twin

CASE ⏚

HTR− ← TH1 battery heater
sheet E16

DC−

INVN · 2×50 mm²

PE1 · 50 mm²

BB4 System − busbar

CHB · 25 mm²

chassis rail bond, via the gland

DCF− → FB1
sheet E3

ORO− ← U2 Orion XS (E4)

MPR− ← U3 MPPT 100/30 (E4)

MPT− ← U4 MPPT 75/15 (E4)

MPV− ← U13 MPPT 100/20 (E4)

DO2− → power-slide pod
sheet E3

BATTERY BANK

- B1P = B2P and B1N = B2N: cut each pair to the longer length (schedule) with identical lugs.
- F1/F2 MRBF 300 A sit directly on each + stud.
- RB1/RB2/SH1/SH2 120 mm² (v0.13): both MRBFs in parallel can deliver 600 A into them.
- Tinned fine-strand, hex-crimped, adhesive heat-shrink; re-torque after 500 km.

ISOLATION

- QS1 opens bank + → system +. BB3 stays live from the chargers: open QP1–QP3, S2 off and ignition off BEFORE opening QS1, so no charger runs without the battery.
- Only F10 (QS1 control) and F11 (TS1 sensor lead) stay live on the bank side.
- S2 off before closing QS1 (MultiPlus inrush).

EARTHING

- DC: one bond only, BB4 → chassis (CHB 25 mm²).
- MultiPlus case → BB4 (PE1, Victron: = DC −).
- AC earth bar → chassis bond (PE2, 6 mm² G/Y), not via the DC negative (E5).
- Orion XS non-isolated: ALT − lands on the body earth, chassis side of the battery sensor (E4).

VICTRON MULTIPLUS-II 12/3000

- F3: Class-T 400 A, holder bolted beside BB3: CT1 link < 200 mm, so F3 is at the source.
- CT1 / INVP / INVN: 2 × 50 mm² per pole; INVP + CT1 ≈ 1.2 m route, through the window.
- MIDI F4–F6 and MEGA F7 on the BB3 studs; F6 40 A serves MPT and MPV (6 mm² each).

LEGEND · line weight ∝ conductor size · ID · size tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC − (black) · Protective earth (green/yellow) · Data (VE.Bus / VE.Direct / USB) · Control (12 V switched) · Signal / sense

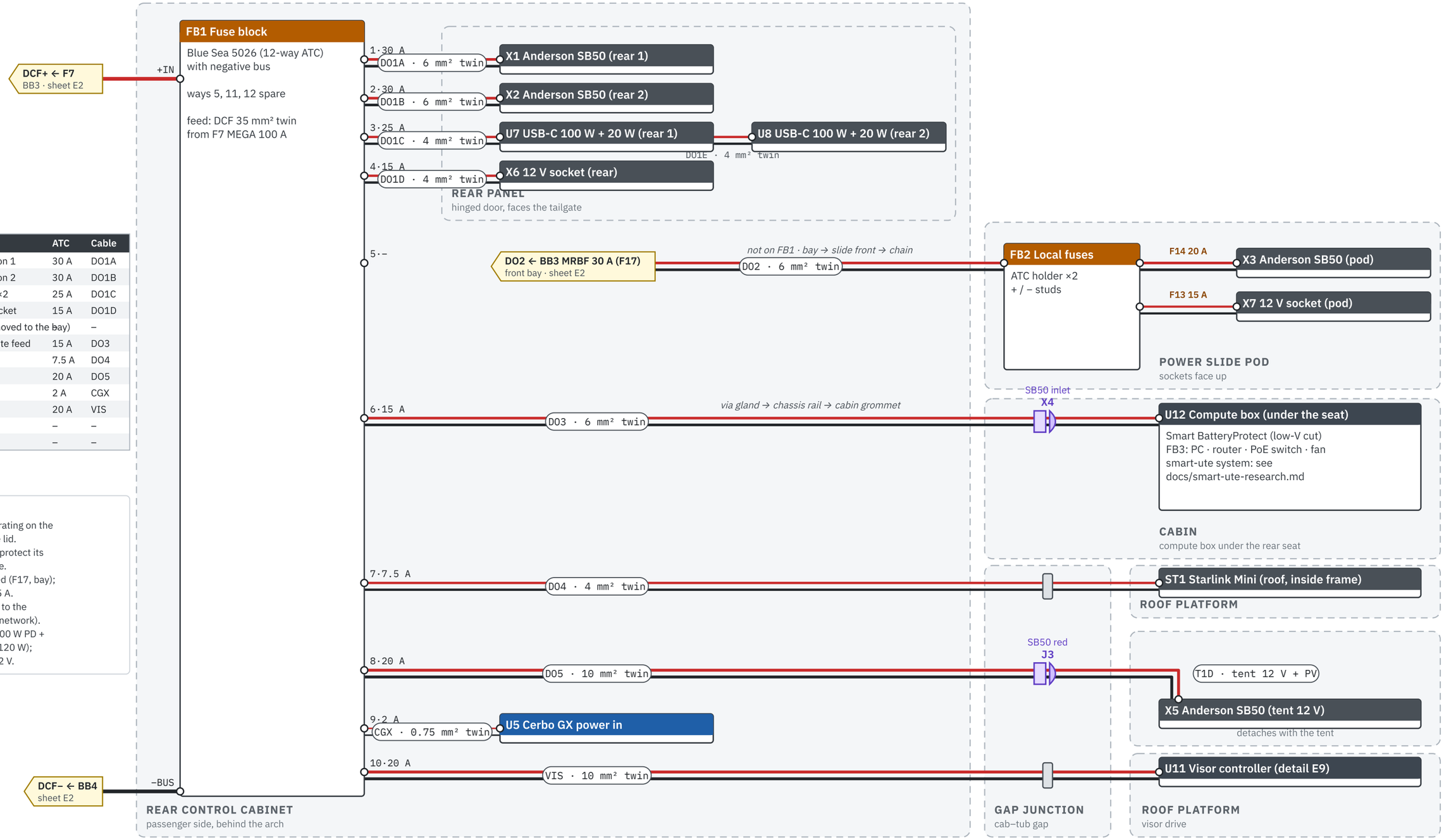
PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E2 of E16**
TITLE **DC power: battery bank and main distribution**	REV / DATE **v0.14 · 2026-10-23**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

DC distribution: fuse block and 12 V outlets

FB1 in the rear control cabinet feeds every 12 V outlet. Twin cables: + from the numbered way, – to the FB1 negative bus.



FB1 WAYS

Way	Circuit	ATC	Cable
1	Rear Anderson 1	30 A	DO1A
2	Rear Anderson 2	30 A	DO1B
3	Rear USB-C ×2	25 A	DO1C
4	Rear 12 V socket	15 A	DO1D
5	spare (pod moved to the bay)		–
6	Cabin compute feed	15 A	DO3
7	Starlink Mini	7.5 A	DO4
8	Tent 12 V	20 A	DO5
9	Cerbo GX	2 A	CGX
10	Visor drive	20 A	VIS
11	spare	–	–
12	spare	–	–

NOTES

- ATC blade fuses, rating on the way label and in the lid.
- Size each fuse to protect its cable, not the device.
- Pod: one 30 A feed (F17, bay); FB2: X3 20 A, X7 15 A.
- Cabin: SB50 inlet to the compute box (PC + network).
- USB-C sockets: 100 W PD + 20 W each (Treklo, 120 W); rear pair ~22 A at 12 V.

LEGEND · line weight ∝ conductor size · `ID · size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) DC – (black)

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E3 of E16**
TITLE **DC distribution: fuse block and 12 V outlets**	REV / DATE **v0.14 · 2026-10-23**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Charging: alternator (DC-DC) and solar

Four independent chargers in the front bay, each output fused at the system busbars (sheet E2).

ENGINE BAY

G1 Alternator
smart / variable

FBX Engine fuse box
ignition add-a-fuse

IGN → F9 2 A → IGN · 1.0 mm²

BT0 Start battery
factory 12 V AGM/EFB

\+ → F8 MIDI 60 A → ALT · 25 mm² twin

\– → *ALT – on the body earth (chassis side of the battery sensor)*

via chassis rail → factory-outlet gland

FRONT BAY: CHARGING BOARD

front wall, above the inverter

U2 DC-DC charger

Victron Orion XS 12/12-50A
non-isolated, 50 A out
input limit 25–30 A (alternator rating unknown)
VE.Direct → Cerbo USB (E6)

IN+ · IN– · H

OUT+ → F4 MIDI 70 A → ORO · 16 mm² twin → **ORO+ → BB3** sheet E2

OUT– → **ORO– → BB4** sheet E2

ROOF PLATFORM

fixed array

PV1 190 W BlueSolar

PV2 130 W BlueSolar

\+ → PVR · 10 mm² → GAP JUNCTION

\– → *parallel via MC4 Y-branch*

PV5 250 W visor
E9

\+ → PVV · 4 mm² chain → MC4 at U11

\– → PVS · 6 mm²

GAP JUNCTION

SB50 grey
J4

TENT ROOF

detachable array

PV3 100 W flex Sunman eArc

PV4 100 W flex Sunman eArc

series link

\+ → T2 · tent PV → J4

\– → PVT · 6 mm²

QP1 PV isolator

PV+ → **U3 MPPT roof**
Victron SmartSolar MPPT 100/30
PV1 + PV2 = 320 W, 30 A out
VE.Direct 2 → Cerbo (E6)

PV– → **U3 MPPT roof**

BAT+ → F5 MIDI 40 A → MPR · 16 mm² twin → **MPR+ → BB3** sheet E2

BAT– → **MPR– → BB4** sheet E2

QP3 PV isolator

PV+ → **U13 MPPT visor**
Victron SmartSolar MPPT 100/20
own charger: panel voltage free
VE.Direct-USB → Cerbo USB 2 (E6)

PV– → **U13 MPPT visor**

BAT+ → F6 shared → MPV · 6 mm² twin → **MPV+ → BB3** sheet E2

BAT– → **MPV– → BB4** sheet E2

QP2 PV isolator

PV+ → **U4 MPPT tent**
Victron SmartSolar MPPT 75/15
PV ≤ 75 V Voc, 15 A out
2 × 100 W in series
VE.Direct 3 → Cerbo (E6)

PV– → **U4 MPPT tent**

BAT+ → F6 MIDI 40 A → MPT · 6 mm² twin → **MPT+ → BB3** sheet E2

BAT– → **MPT– → BB4** sheet E2

CHARGER SETTINGS

- Absorption / float / no temperature compensation set to iTechworld's $LiFePO_4$ figures in all four chargers (the BMS has no Victron link).
- Orion XS: charger mode, input limit 25–30 A until the alternator rating is known; IGN on H. Two panels in parallel need no string fuses.
- DVCC on, SmartShunt as the battery monitor (Venus OS ≥ 3.20). QP1–QP3: 2-pole DC isolators; open them before QS1 (sheet E2).

LEGEND · line weight ∝ conductor size · `ID` · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC – (black) · PV + · PV – · Signal / sense · Factory vehicle wiring

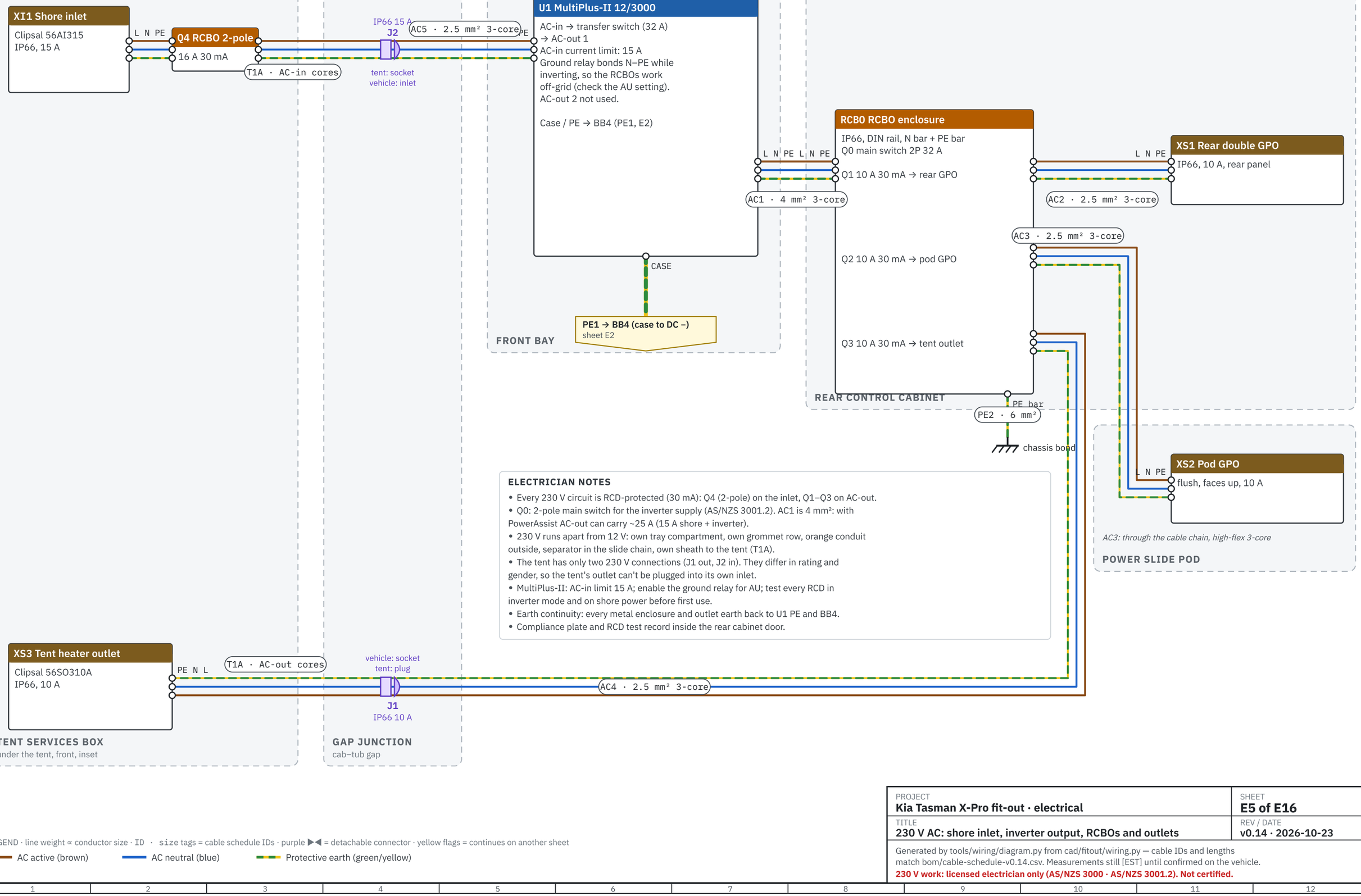
PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E4 of E16**
TITLE **Charging: alternator (DC-DC) and solar**	REV / DATE **v0.14 · 2026-10-23**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

230 V AC: shore inlet, inverter output, RCBOs and outlets

Licensed electrician to install and certify (AS/NZS 3000, AS/NZS 3001.2). Active brown · neutral blue · earth green/yellow.



ELECTRICIAN NOTES

- Every 230 V circuit is RCD-protected (30 mA): Q4 (2-pole) on the inlet, Q1–Q3 on AC-out.
- Q0: 2-pole main switch for the inverter supply (AS/NZS 3001.2). AC1 is 4 mm²: with PowerAssist AC-out can carry ~25 A (15 A shore + inverter).
- 230 V runs apart from 12 V: own tray compartment, own grommet row, orange conduit outside, separator in the slide chain, own sheath to the tent (T1A).
- The tent has only two 230 V connections (J1 out, J2 in). They differ in rating and gender, so the tent's outlet can't be plugged into its own inlet.
- MultiPlus-II: AC-in limit 15 A; enable the ground relay for AU; test every RCD in inverter mode and on shore power before first use.
- Earth continuity: every metal enclosure and outlet earth back to U1 PE and BB4.
- Compliance plate and RCD test record inside the rear cabinet door.

LEGEND · line weight ∝ conductor size · ID · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

AC active (brown) · AC neutral (blue) · Protective earth (green/yellow)

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E5 of E16**
TITLE **230 V AC: shore inlet, inverter output, RCBOs and outlets**	REV / DATE **v0.14 · 2026-10-23**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Control and communications

Cerbo GX gathers every Victron device for the GX Touch, Home Assistant and VRM. Hard-wired switches at the rear panel.

MONITORING

Cerbo GX in the rear cabinet

ST1 Starlink Mini
Wi-Fi router / uplink

Wi-Fi (VRM uplink, optional)

Victron integration · Modbus-TCP / MQTT

HA Home Assistant

Wi-Fi

U5 Cerbo GX MK2
Victron GX device, Venus OS ≥ 3.20
3 × VE.Direct, 2 × VE.Bus (RJ45)
3 × USB, HDMI, Ethernet, Wi-Fi, BLE
VE.Can 1/2, relays, tank / temp / digital inputs: unused

power: FB1 way 9, 2 A (CGX)

front bay / charging board

U6 GX Touch 50
5" touch display, rear panel

HDMI — D3 · HDMI+USB — USB 3

CGX ← FB1 way 9 (2 A)
sheet E3

PWR

VE.Bus	D1 · VE.Bus RJ45	VE.Bus	**U1 MultiPlus-II**
VE.Direct 1	D2 · VE.Direct	VE.Direct	**P1 SmartShunt 500 A**
VE.Direct 2	D2 · VE.Direct	VE.Direct	**U3 MPPT 100/30 (roof)**
VE.Direct 3	D2 · VE.Direct	VE.Direct	**U4 MPPT 75/15 (tent)**
USB 1	D4 · VE.Direct-USB	VE.Direct	**U2 Orion XS 12/12-50A**
USB 2	D5 · VE.Direct-USB	VE.Direct	**U13 MPPT 100/20 (visor)**

RBF ← BB1 via F10 2 A
bank side, always live · E2

S1 Battery isolate switch
Blue Sea 2155 Contura
SPDT ON-ON with LED
rear panel

+12V			
CTRL	SW1 · 4-core 1.0 mm²	red	
LED		yel	
GND		blk	

QS1 ML-RBS 7713 control harness
red: CONTROL (high = close)
yellow: LED OUTPUT
black: GROUND
verify colours vs Blue Sea doc

→ BB4 system –
sheet E2

S2 Inverter on/off
SPST rocker, rear panel

1	SW2 · 2-core 0.75 mm²	L	
2		H	

U1 MultiPlus-II remote on/off
remove the factory L–H bridge

S3 Visor deploy / stow
momentary rocker + LED, rear panel

DEP		
STOW	VSW · 4-core 0.75 mm²	
LED		
COM		

VSW (4-core) → U11 visor controller
roof · sheet E9

IGN ← F9 2 A (E4) — H — **U2 Orion XS remote on/off**
H ← ignition 12 V (IGN)
remove the L–H bridge

VIG: ignition → U11 visor interlock
sheet E9

- S1 drives QS1 open/closed; QS1 latches, so S1 position is the command and the LED confirms the contacts.
- S2 open = MultiPlus fully off (inverter and charger).

Modes (on / charger only) are also on the GX Touch.

CONTROL CIRCUITS

switches on the rear panel

LEGEND · line weight ∝ conductor size · `ID` · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC – (black) · Data (VE.Bus / VE.Direct / USB) · Control (12 V switched) · Signal / sense · Wireless (Wi-Fi / BLE)

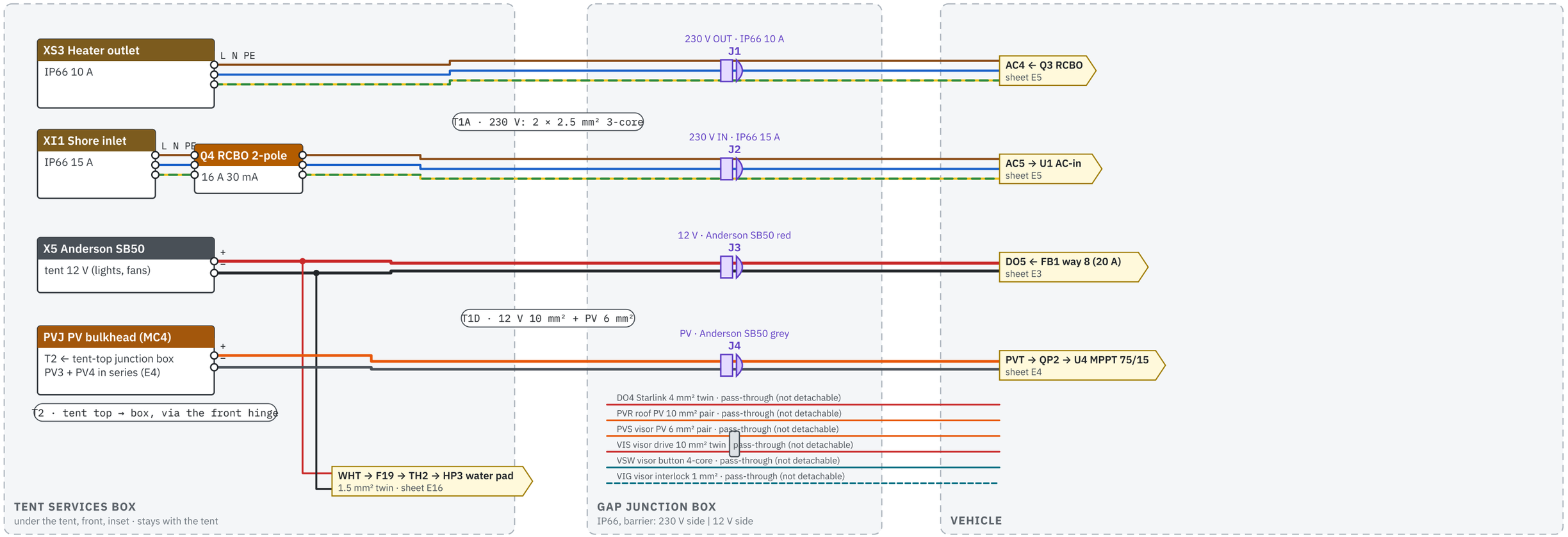
PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E6 of E16**
TITLE **Control and communications**	REV / DATE **v0.14 · 2026-10-23**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Detachable tent harness and gap junction

Four keyed connectors at the cab–tub gap. Unplug J1–J4, cap them, and the tent lifts off with its harness and services box.



J1 · 230 V OUT TO TENT · IP66 10 A (CLIPSAL 56 SERIES)

Pin	Function	Conductor	Vehicle side	Tent side
A	Active	2.5 mm² brown	socket (live)	plug
N	Neutral	2.5 mm² blue		
E	Earth	2.5 mm² green/yellow		

J2 · 230 V SHORE IN · IP66 15 A INLET + EXTENSION SOCKET

Pin	Function	Conductor	Vehicle side	Tent side
A	Active (shore)	2.5 mm² brown	appliance inlet	socket (live)
N	Neutral	2.5 mm² blue		
E	Earth	2.5 mm² green/yellow		

J3 · 12 V · ANDERSON SB50 RED

Pin	Function	Conductor
+	12 V +	10 mm² red
–	12 V –	10 mm² black

J4 · PV · ANDERSON SB50 GREY

Pin	Function	Conductor
+	PV +	6 mm² solar
–	PV –	6 mm² solar

DETACH / REFIT

1. Switch the inverter off (S2) and isolate shore power at the source.
2. Unplug J2 (shore in) first, then J1, J3, J4.
3. Fit the caps to all eight halves (IP66 caps / SB50 dust covers).
4. Unclip T1A and T1D from the lid-front saddles, then lift the tent.

Refit in reverse. SB50 colours are keyed: red and grey won't mate, and J1/J2 differ in rating and gender, so nothing can cross-connect.

LEGEND · line weight ∝ conductor size · `ID` · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC – (black) · PV + · PV – · AC active (brown) · AC neutral (blue) · Protective earth (green/yellow) · Control (12 V switched) · Signal / sense

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E7 of E16**
TITLE **Detachable tent harness and gap junction**	REV / DATE **v0.14 · 2026-10-23**

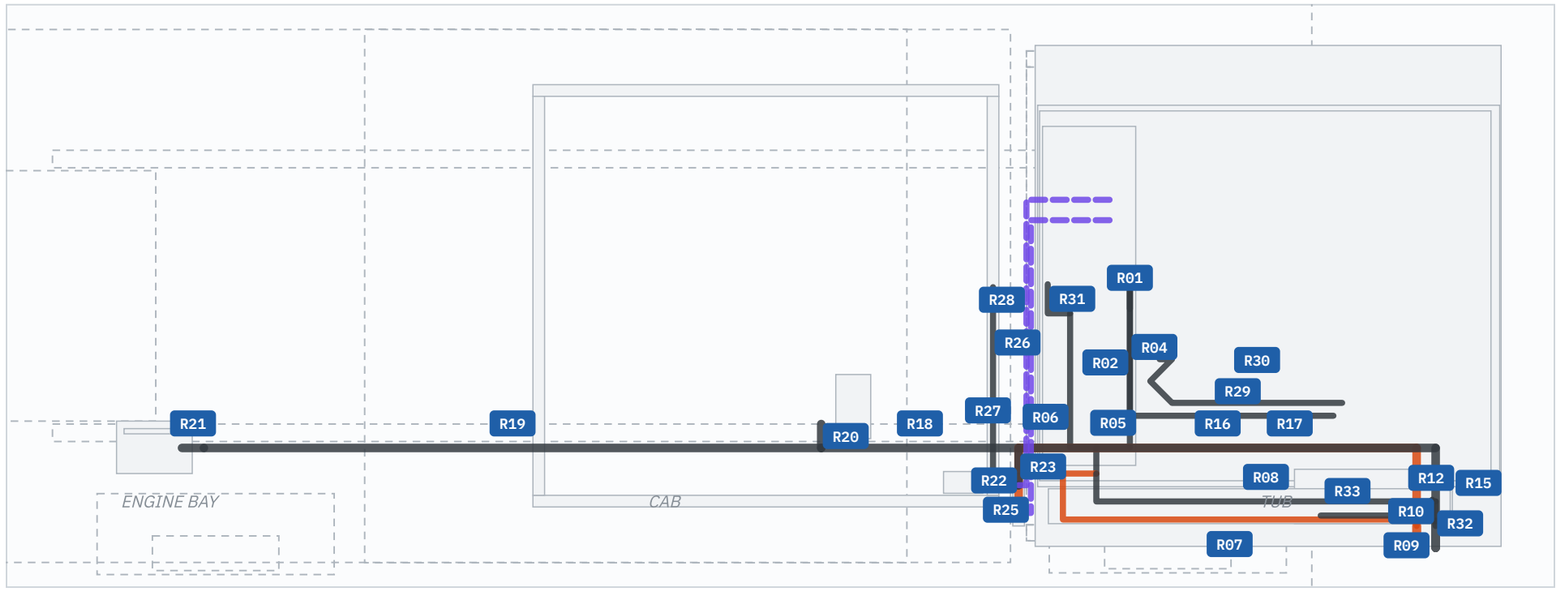
Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

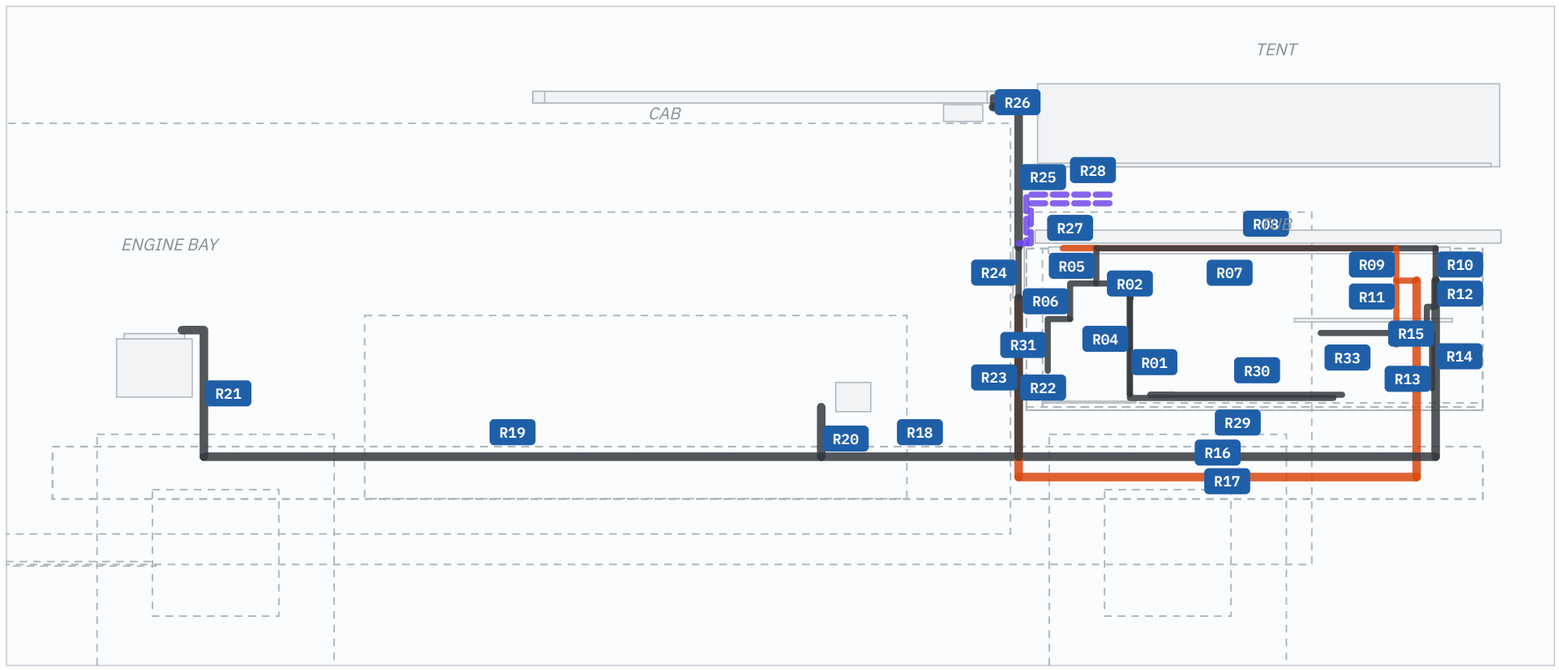
Cable routes: route map and run schedule

Every physical run (R-code) from the routed 3D model. Detail by zone on E10–E14; bundle sections and fill on E15.

PLAN — front to the left, driver side up · 1:21 on A3



SIDE — from the passenger side · 1:21 on A3



12 V + / DC · DC − · 230 V (sheathed flex) · PV · data / control · earth · tent harness (detachable) · R07 run code (schedule on E8) · P-clip / saddle

RUN SCHEDULE

Run	Name	m	n	Containment	Fill	V
R01	Bay riser, busbar side	0.38	11	ties / mounts		mixed
R02	Bay rear channel, high	0.58	9	ties / mounts		mixed
R03	Bay rear channel, cross link	0.11	1	ties / mounts		12 V
R04	Bay rear channel, low (to the slid	0.85	2	ties / mounts		mixed
R05	Bay exit over the inverter	0.33	17	ties / mounts		mixed
R31	Bay front channel (chargers to the	0.35	3	ties / mounts		12 V
R06	Charging-board loom	0.67	9	ties / mounts		12 V
R07	Tray, 230 V compartment	1.42	3	tray		230 V
R08	Tray, 12 V compartment	1.26	14	tray		12 V
R09	Wall channel, 230 V drop	0.20	4	wall channel		230 V
R10	Wall channel, 12 V drop	0.18	14	wall channel		12 V
R32	Gland plate, 12 V rows	0.02	13	ties / mounts		12 V
R11	Cabinet, 230 V entry	0.22	3	ties / mounts		230 V
R12	Cabinet entry (12 V slot)	0.22	13	ties / mounts		12 V
R33	Cabinet top loom	0.36	15	ties / mounts		12 V
R13	Rear-panel loom	0.26	8	ties / mounts		12 V
R14	Wall cavity + under floor, 230 V	1.09	2	orange conduit 25	35%	230 V
R15	Wall cavity + under floor, 12 V	1.02	13	split conduit 50	40%	12 V
R16	Chassis rail, 230 V, tub to the ga	1.37	2	orange conduit 25	35%	230 V
R17	Chassis rail, 12 V, tub to the gap	1.43	11	split conduit 50	34%	12 V
R18	Chassis rail, 12 V, gap to the cab	0.68	3	split conduit 25	45%	12 V
R19	Chassis rail, 12 V, cabin to the e	2.12	2	split conduit 25	36%	12 V
R20	Cabin branch	0.25	1	split conduit 13	35%	12 V
R21	Engine-bay riser	0.51	2	split conduit 25	36%	12 V
R22	Gap riser, 230 V	0.84	2	orange conduit 25	35%	230 V
R23	Gap riser, 12 V	0.67	8	split conduit 40	36%	12 V
R24	Junction box pass-through	0.17	6	ties / mounts		12 V
R25	Cab back riser to the roof	0.57	6	split conduit 32	39%	12 V
R26	Platform rear end member	0.71	2	in extrusion		12 V
R27	Tent harness, 230 V	1.45	1	spiral wrap 25	35%	230 V
R28	Tent harness, 12 V + PV	1.46	1	spiral wrap 25	44%	12 V
R29	Slide trunk (fixed, on the glide t	0.71	2	ties / mounts		mixed
R30	Power-slide cable chain	0.84	2	drag chain		mixed

ROUTING RULES

- 230 V cables (orange runs) never share a conduit, gland row or tray compartment with 12 V: own grommet row, own orange conduit, separator in the slide chain.
- Outside the tub every run is in split conduit on rubber-lined P-clips every 300 mm (circles on E12), into M6 rivnuts - no welding or drilling the rail flanges.
- Fill = cable area / conduit area (≤ 50 % for split conduit laid in).
- Lengths: routed in the model; schedule adds 10 % + 0.3 m per cable.

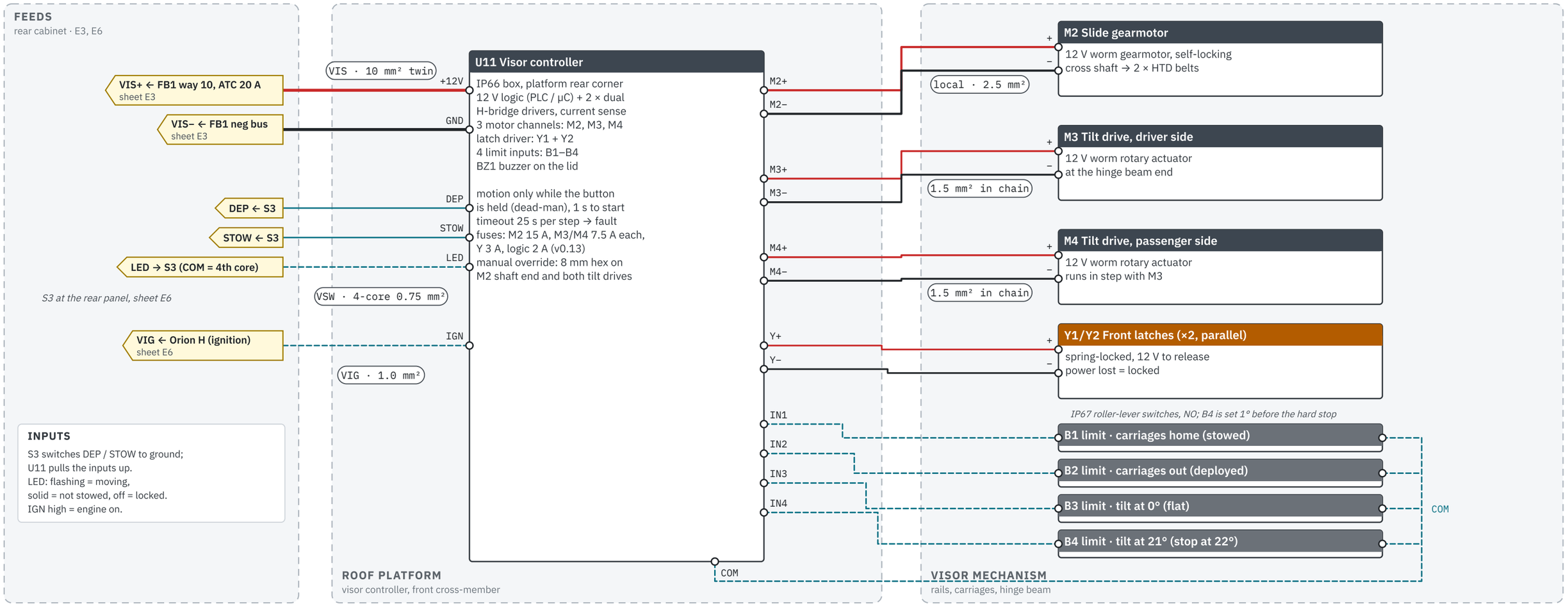
PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E8 of E16**
TITLE **Cable routes: route map and run schedule**	REV / DATE **v0.14 · 2026-10-23**

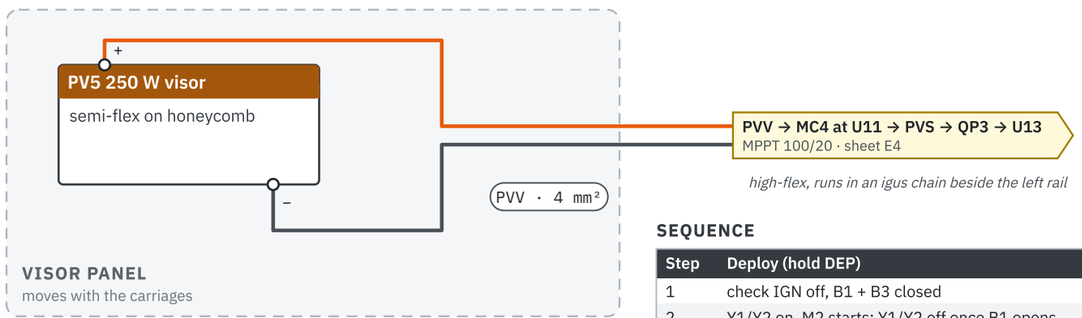
Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Visor drive: slide, tilt and locking

The windscreen visor panel (PV5) slides out on two rails and tilts to 20°. One button at the rear panel; every motion is interlocked.





SEQUENCE

Step	Deploy (hold DEP)	Stow (hold STOW)
1	check IGN off, B1 + B3 closed	allowed any time (IGN on too)
2	Y1/Y2 on, M2 starts; Y1/Y2 off once B1 opens	M3 + M4 tilt down until B3
3	M2 extends until B2	M2 retracts to B1; pins spring into ramped strikers
4	M3 + M4 tilt up to 20°, B4 = limit	latches need no power to lock
5	worms hold the panel, nothing powered	LED off = locked for travel

INTERLOCKS AND SAFETY

- No deploy with the ignition on. Engine on while not stowed → BZ1 sounds, LED flashes; nothing moves without the button.
- M2 runs only with B3 closed: the panel slides flat, then tilts.
- Over-current on any channel → stop and back off 2 mm (pinch).
- Mechanical stop at 22° keeps ≥ 87 mm to the glass even if B4 fails; worm drives cannot be back-driven by wind or load.
- Latches fail locked. Stow when gusts exceed ~40 km/h.
- Y1/Y2 are short-duty (≈5 s): energised < 3 s per deploy.
- U11 draws < 20 mA idle; isolate with FB1 way 10.

LEGEND · line weight ∝ conductor size · `ID` · `size` tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC − (black) · PV + · PV − · Control (12 V switched) · Signal / sense

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E9 of E16**
TITLE **Visor drive: slide, tilt and locking**	REV / DATE **v0.14 · 2026-10-23**

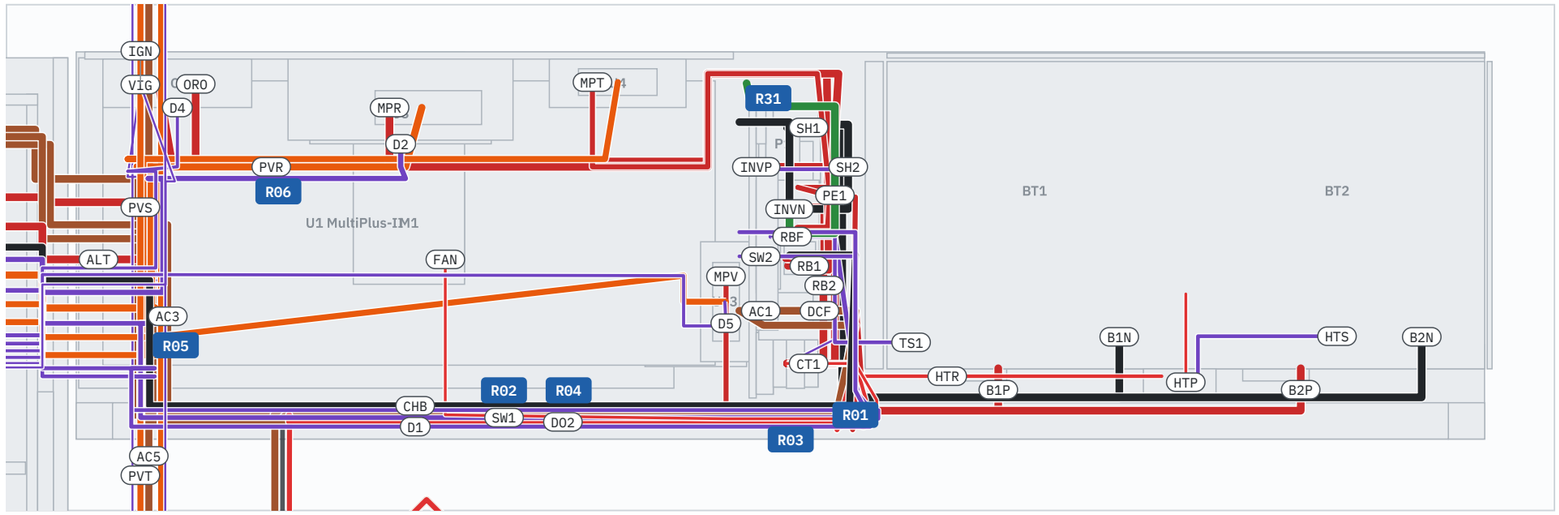
Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

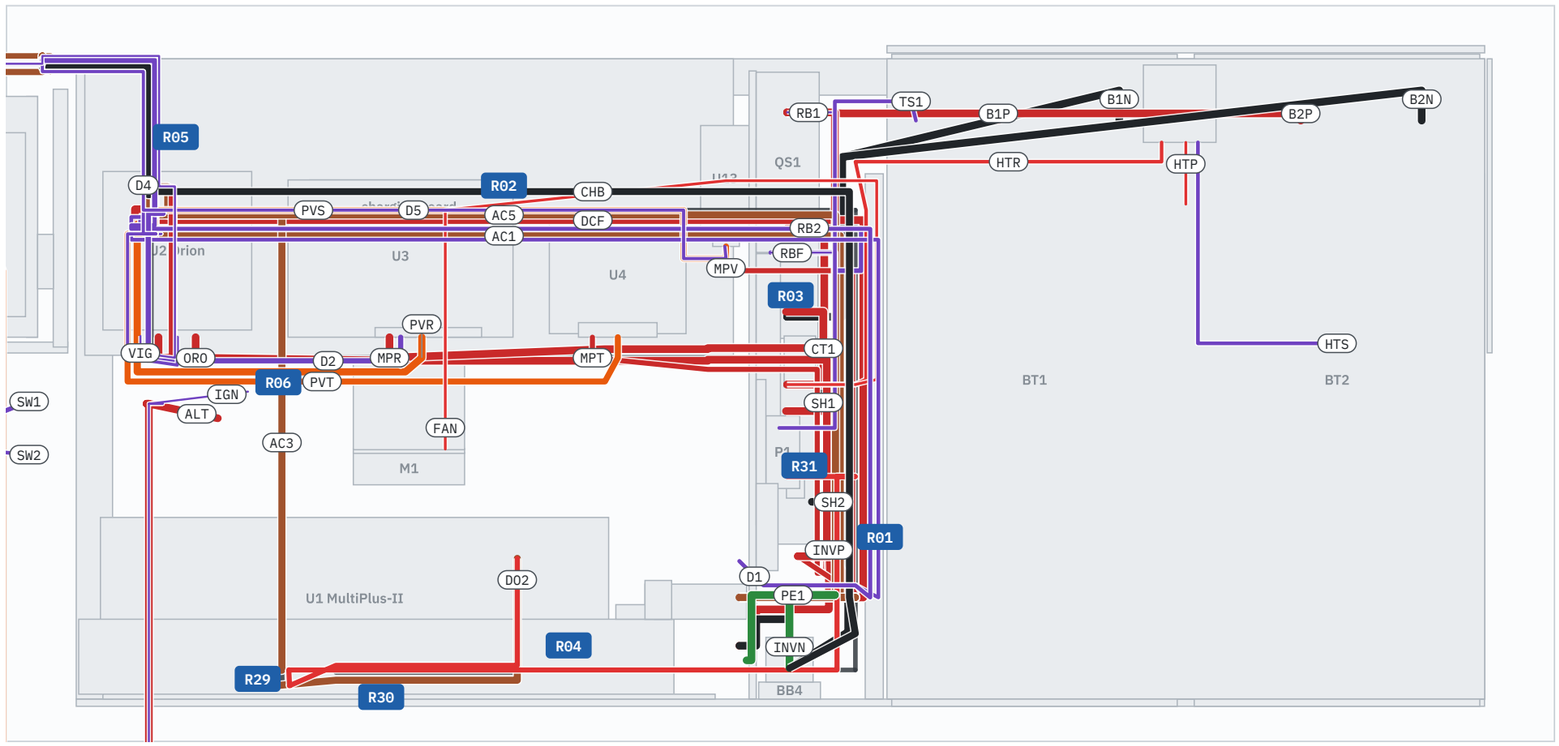
Cable routes: front bay

Busbar side, rear channel (behind the board edge and the inverter), front channel, charging-board loom and the exits to the tray and the slide.

PLAN — front wall at the top, driver side to the right · 1:5 on A3



VIEW FROM THE REAR — looking forward (guard removed) · 1:5 on A3



CABLES IN THE BAY

ID	From	To	Runs here
AC1	MultiPlus-II AC-out 1	RCBO enclosure (Q0 main switch)	R01 R02 R05
AC3	RCBO Q2	Power-slide pod GPO	R05 R02 R04
AC5	Gap junction J2 (from the tent's …	MultiPlus-II AC-in	R05 R02 R01
ALT	Start battery + (F8 MIDI 60 A); -…	Orion XS input	R05 R06
B1N	Battery 1 -	Bank - busbar	local
B1P	Battery 1 + (MRBF 300 A)	Bank + busbar	local
B2N	Battery 2 -	Bank - busbar	local
B2P	Battery 2 + (MRBF 300 A)	Bank + busbar	local
CHB	System - busbar	Chassis bond point (passenger rai…	R01 R02 R05
CT1	System + busbar	Class-T 400 A	local
D1	Cerbo GX	MultiPlus-II	R05 R02 R01
D2	Cerbo GX VE.Direct 1-3	SmartShunt / MPPT 100/30 / MPPT 7…	R05 R06
D4	Cerbo GX USB 1	Orion XS VE.Direct	R05 R06
D5	Cerbo GX USB 2	MPPT 100/20 (visor) VE.Direct	R05
DCF	System + busbar (MEGA 100 A)	Rear DC fuse block	R01 R02 R05
DO2	System + busbar spare stud (MRBF …	Power-slide pod (FB2)	R01 R04
FAN	System + busbar (inline 5 A)	92 mm thermostatic fan	R01 R02
HTP	Heater controller TH1 (relay out)	Heater pads HP1 + HP2 (between th…	local
HTR	System + busbar (F18 10 A inline)…	Heater controller TH1	R01
HTS	Heater controller TH1 (sensor in)	NTC probe on battery 2 (lid face,…	local
IGN	Engine-bay fuse box (add-a-fuse, …	Orion XS remote H	R05 R06
INVN	System - busbar	MultiPlus-II DC -	local
INVP	Class-T 400 A	MultiPlus-II DC +	local
MPR	MPPT 100/30 (roof)	System +/- busbars (MIDI fuse)	R06 R31
MPT	MPPT 75/15 (tent)	System +/- busbars (MIDI fuse)	R31
MPV	MPPT 100/20 (visor)	System +/- busbars (MIDI fuse)	R03 R01
ORO	Orion XS 12/12-50A	System +/- busbars (MIDI fuse)	R06 R31
PE1	MultiPlus-II chassis earth stud	System - busbar	local
PVR	Roof panels PV1 + PV2 (parallel, …	QP1 -> MPPT 100/30 PV in	R05 R06
PVS	U11 box (MC4 couplers at the viso…	QP3 -> MPPT 100/20 (visor) PV in	R05
PVT	Gap junction J4 (from the tent pa…	QP2 -> MPPT 75/15 PV in	R05 R06
RB1	Bank + busbar	Remote battery switch IN	local
RB2	Remote battery switch OUT	System + busbar	local
RBF	Bank + busbar	ML-RBS control feed (inline 2 A)	local
SH1	Bank - busbar	SmartShunt (battery side)	local
SH2	SmartShunt (system side)	System - busbar	local
SW1	Rear panel ML-RBS switch (Blue Se…	ML-RBS control harness	R05 R02 R01
SW2	Rear panel switch S2	MultiPlus-II remote terminals	R05 R02 R01
TS1	Battery 1 + post (lug with sensor…	SmartShunt Vbatt+ and Aux	local
VIG	Orion XS remote H (ignition, F9)	Visor controller interlock input	R06 R05

12 V + / DC · DC − · 230 V (sheathed flex) · PV · data / control · earth · tent harness (detachable) · R07 run code (schedule on E8) · P-clip / saddle

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E10 of E16**
TITLE **Cable routes: front bay**	REV / DATE **v0.14 · 2026-10-23**

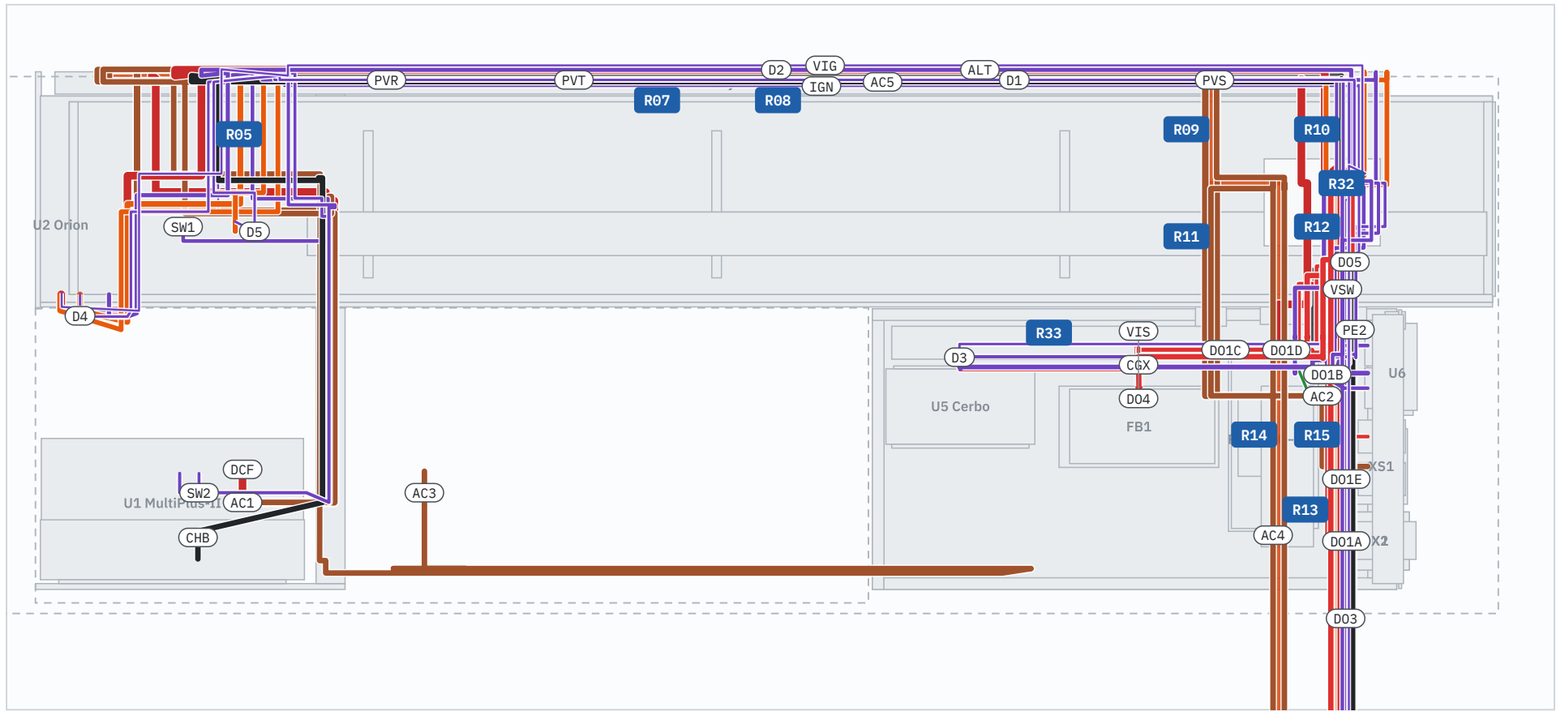
Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

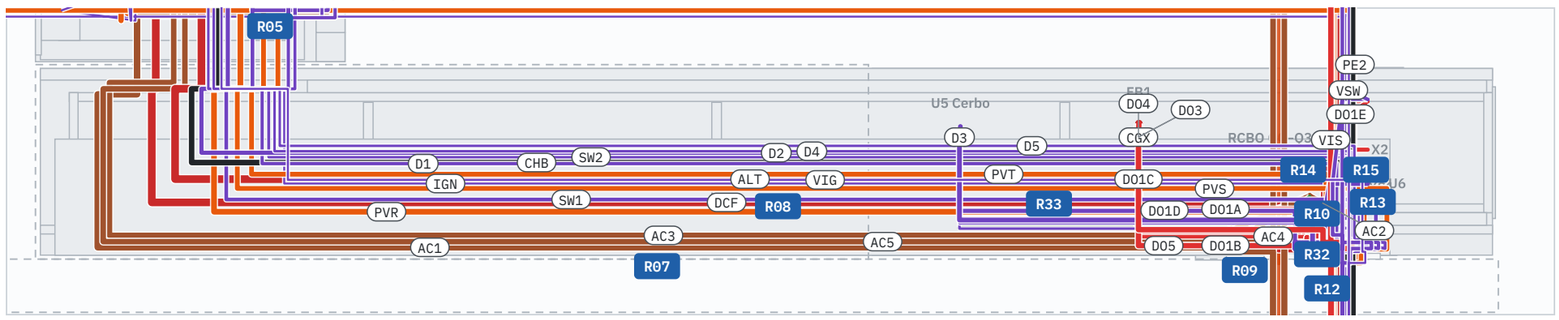
Cable routes: tray, wall channel, gland plate and rear cabinet

Passenger side of the tub. 230 V (orange) on the wall side of the tray divider, its own drop, gland row and cabinet gland.

VIEW OUTBOARD — from inside the tub toward the passenger wall · 1:6 on A3



PLAN — passenger strip (tray, channel, cabinet) · 1:6 on A3



CABLES IN THE TRAY, CHANNEL AND CABINET

ID	From	To	Runs here
AC1	MultiPlus-II AC-out 1	RCBO enclosure (Q0 main switch)	R07 R09 R11
AC2	RCBO Q1	Rear panel double GPO	local
AC3	RCBO Q2	Power-slide pod GPO	R11 R09 R07
AC4	RCBO Q3	Gap junction J1 -> tent services ...	R11 R09
AC5	Gap junction J2 (from the tent's ...	MultiPlus-II AC-in	R09 R07
ALT	Start battery + (F8 MIDI 60 A); -...	Orion XS input	R32 R10 R08
CGX	DC fuse block	Cerbo GX power in	R33
CHB	System - busbar	Chassis bond point (passenger rai...	R08 R10 R32
D1	Cerbo GX	MultiPlus-II	R33 R12 R10 R08
D2	Cerbo GX VE.Direct 1-3	SmartShunt / MPPT 100/30 / MPPT 7...	R33 R12 R10 R08
D3	Cerbo GX (HDMI + USB)	GX Touch 50	R33 R13
D4	Cerbo GX USB 1	Orion XS VE.Direct	R33 R12 R10 R08
D5	Cerbo GX USB 2	MPPT 100/20 (visor) VE.Direct	R33 R12 R10 R08
DCF	System + busbar (MEGA 100 A)	Rear DC fuse block	R08 R10 R12 R33
DO1A	DC fuse block	Rear panel Anderson 1	R33 R13
DO1B	DC fuse block	Rear panel Anderson 2	R33 R13
DO1C	DC fuse block	Rear panel USB-C x2	R33 R13
DO1D	DC fuse block	Rear panel 12 V socket	R33 R13
DO1E	USB-C U7 terminals	USB-C U8 terminals	local
DO3	DC fuse block	Compute box SB50 inlet (under the...	R33 R12 R32
DO4	DC fuse block	Starlink Mini (roof)	R33 R12 R32
DO5	DC fuse block	Gap junction J3 -> tent (SB50 red)	R33 R12 R32
IGN	Engine-bay fuse box (add-a-fuse, ...	Orion XS remote H	R32 R10 R08
PE2	RCBO enclosure earth bar	Chassis bond point (passenger rai...	R12 R32
PVR	Roof panels PV1 + PV2 (parallel, ...	QP1 -> MPPT 100/30 PV in	R32 R10 R08
PVS	U11 box (MC4 couplers at the viso...	QP3 -> MPPT 100/20 (visor) PV in	R32 R10 R08
PVT	Gap junction J4 (from the tent pa...	QP2 -> MPPT 75/15 PV in	R32 R10 R08
SW1	Rear panel ML-RBS switch (Blue Se...	ML-RBS control harness	R13 R12 R10 R08
SW2	Rear panel switch S2	MultiPlus-II remote terminals	R13 R12 R10 R08
VIG	Orion XS remote H (ignition, F9)	Visor controller interlock input	R08 R10 R32
VIS	DC fuse block (way 10)	Visor controller U11 (roof)	R33 R12 R32
VSW	Rear panel visor button (S3)	Visor controller U11 (roof)	R13 R12 R32

FIXING

- Tray (R07/R08): 120 mm aluminium over the drawer carcass; 230 V outboard of the divider, 12 V inboard; ties through the slots every 120 mm.
- Wall channel (R09/R10): 22 mm behind the drawer runner panel; two layers of 12 V cables, 230 V in its own drop 135 mm further forward.
- Gland plate (old 240 V outlet hole, 120 × 90): one grommet row for 230 V (R09), two rows for 12 V (R32); cut the plate from 3 mm aluminium.
- Cabinet: 12 V through a 100 × 22 slot with a split grommet (R12), 230 V through its own 28 mm gland (R11) beside the RCBO enclosure.

12 V + / DC · DC – · 230 V (sheathed flex) · PV · data / control · earth · tent harness (detachable) · R07 run code (schedule on E8) · P-clip / saddle

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E11 of E16**
TITLE **Cable routes: tray, wall channel, gland plate and rear cabinet**	REV / DATE **v0.14 · 2026-10-23**

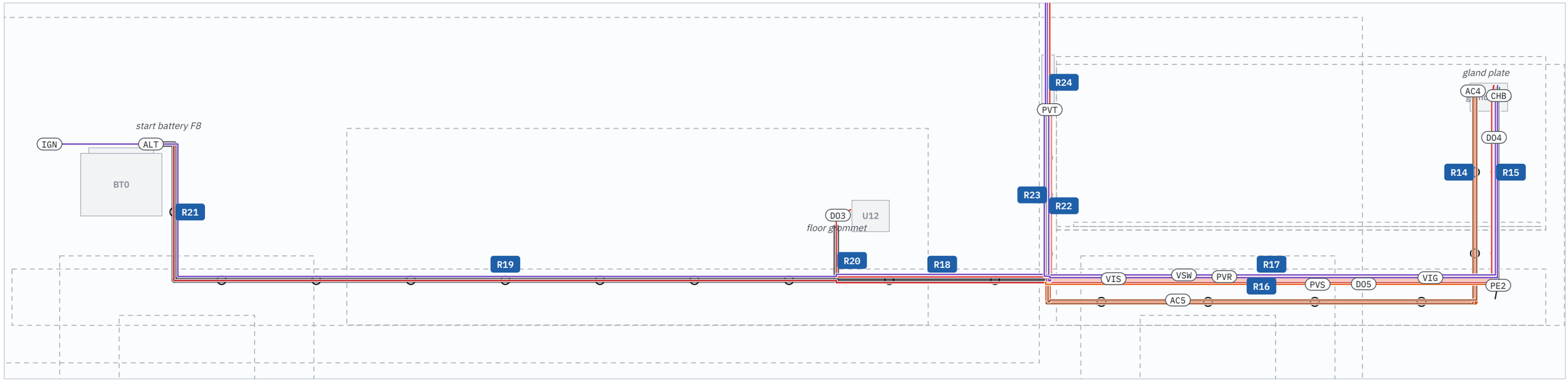
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230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

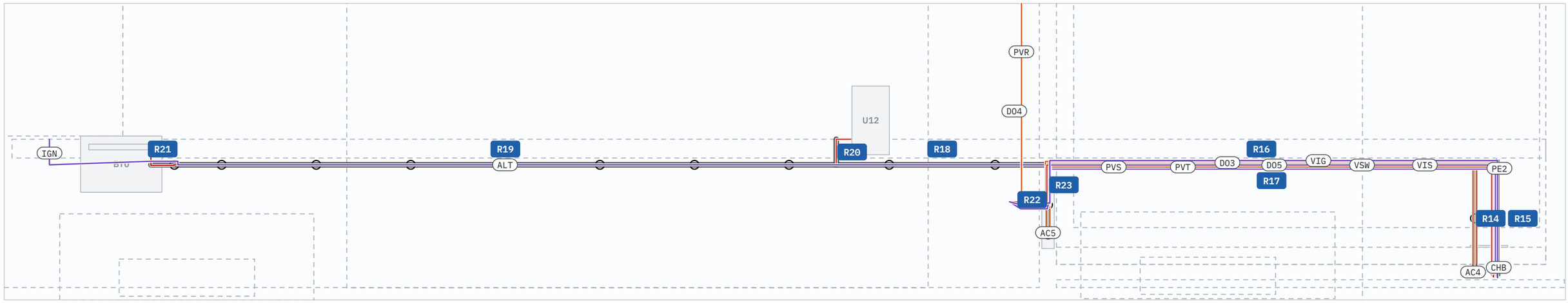
Cable routes: wall cavity, chassis rail, cabin and engine bay

Outside the tub: 12 V in black split conduit, 230 V in orange conduit below it on the rail web; P-clips every 300 mm.

SIDE — from the passenger side · 1:13 on A3



PLAN — passenger half, from above · 1:13 on A3



OUTSIDE RUNS

Run	Name	m	Conduit	Fill	Clips	Cables
R14	Wall cavity + under floor, 230 V	1.09	25 mm	35%	2	AC4 AC5
R15	Wall cavity + under floor, 12 V	1.02	50 mm	40%	2	PE2 DO3 DO4 DO5 ALT IGN PVR PVT PVS CHB VIS VSW VIG
R16	Chassis rail, 230 V, tub to the gap	1.37	25 mm	35%	4	AC4 AC5
R17	Chassis rail, 12 V, tub to the gap	1.43	50 mm	34%	4	DO3 DO4 DO5 ALT IGN PVR PVT PVS VIS VSW VIG
R18	Chassis rail, 12 V, gap to the cabin	0.68	25 mm	45%	2	DO3 ALT IGN
R19	Chassis rail, 12 V, cabin to the engine	2.12	25 mm	36%	7	ALT IGN
R20	Cabin branch	0.25	13 mm	35%	0	DO3
R21	Engine-bay riser	0.51	25 mm	36%	1	ALT IGN
R22	Gap riser, 230 V	0.84	25 mm	35%	2	AC4 AC5
R23	Gap riser, 12 V	0.67	40 mm	36%	1	DO4 DO5 PVR PVT PVS VIS VSW VIG

12 V + / DC · DC – · 230 V (sheathed flex) · PV · data / control · earth · tent harness (detachable) · R07 run code (schedule on E8) · P-clip / saddle

FIXING

- Wall cavity (between the tub liner and the outer skin): both conduits drop from the gland plate and leave through the cavity's bottom drain opening - check on the vehicle (measurement sheet).
- Rail: conduits on the outer face of the web, 12 V above 230 V; keep 150 mm from the exhaust and use heat-rated conduit near it.
- Cabin: one sealed grommet in the floor under the rear seat.
- Engine bay: up the inner guard to the start battery (F8 within 150 mm).

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E12 of E16**
TITLE **Cable routes: wall cavity, chassis rail, cabin and engine bay**	REV / DATE **v0.14 · 2026-10-23**

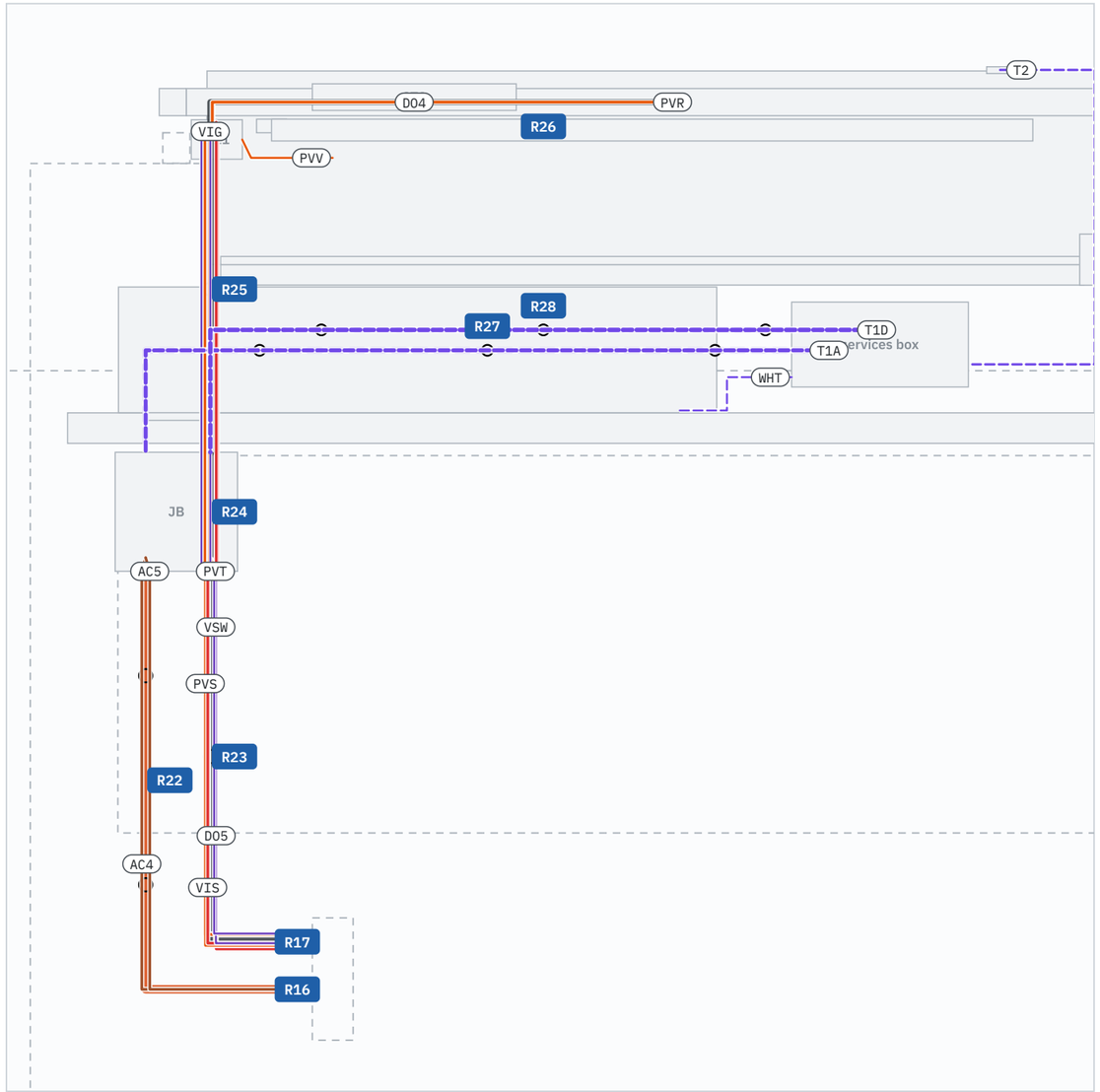
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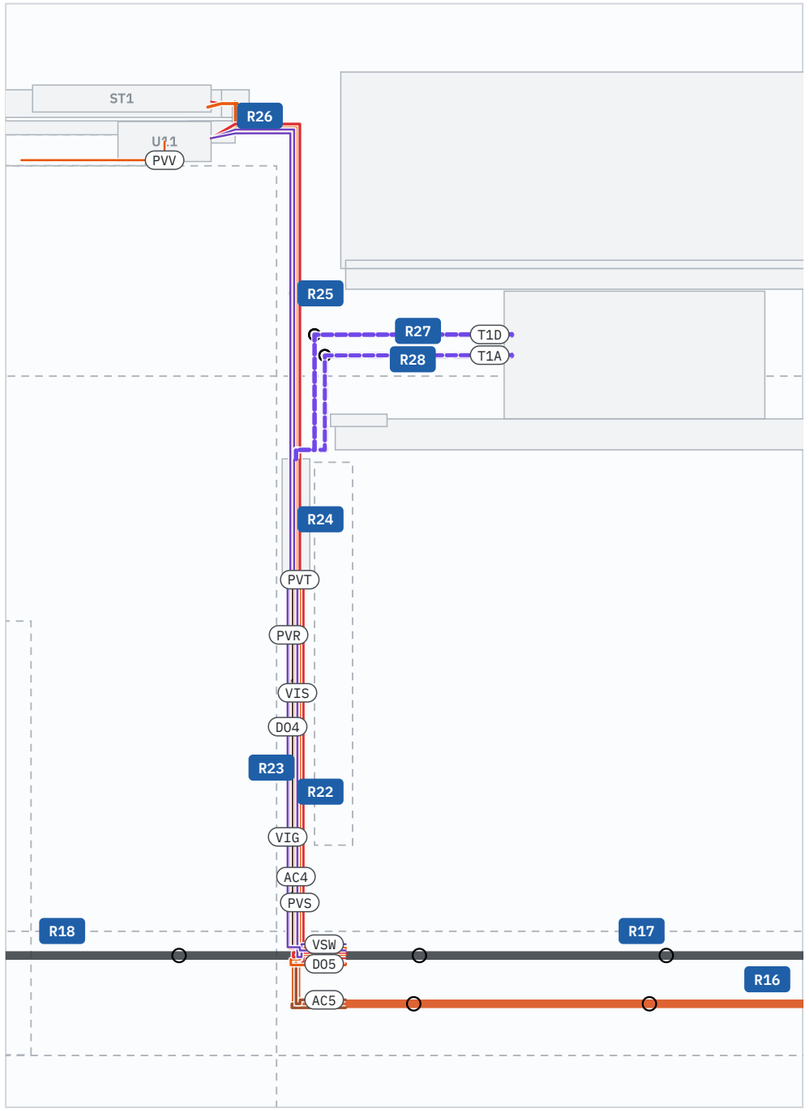
Cable routes: cab–tub gap, junction box, roof and tent harnesses

Risers up the gap to the junction box (230 V and 12 V sides), on up the cab back to the platform's hollow rear end member; tent harnesses across the lid front.

VIEW FORWARD — at the cab back (tent and lid shown dashed) · 1:9 on A3



SIDE — from the passenger side · 1:9 on A3



CABLES HERE

ID	From	To	Runs here
AC4	RCBO Q3	Gap junction J1 ->...	R22
AC5	Gap junction J2 (f...	MultiPlus-II AC-in	R22
DO4	DC fuse block	Starlink Mini (roo...	R23 R24 R25 R26
DO5	DC fuse block	Gap junction J3 ->...	R23
PVR	Roof panels PV1 + ...	QP1 -> MPPT 100/30...	R26 R25 R24 R23
PVS	U11 box (MC4 coupl...	QP3 -> MPPT 100/20...	R25 R24 R23
PVT	Gap junction J4 (f...	QP2 -> MPPT 75/15 ...	R23
PVV	Visor panel juncti...	U11 box (MC4 coupl...	local
T1A	Tent services box ...	Gap junction J1 + ...	R27
T1D	Tent services box ...	Gap junction J3 + ...	R28
T2	Tent-top junction ...	Services box PV bu...	local
VIG	Orion XS remote H ...	Visor controller i...	R23 R24 R25
VIS	DC fuse block (way...	Visor controller U...	R23 R24 R25
VSW	Rear panel visor b...	Visor controller U...	R23 R24 R25
WHT	Tent services box ...	Water can pad HP3 ...	local

FIXING

- Junction box (JB): internal barrier; J1/J2 and the 230 V conduit (R22) on the outboard side, J3/J4 and the 12 V conduit (R23) inboard.

Roof cables pass straight through on the 12 V side (R24) and are not detachable.

- Cab back (R25): 32 mm conduit on stick-on clamps up the C-pillar corner, over the roof edge under the platform, into the end member.
- End member (R26): hollow 34 × 34 extrusion; grommets where cables enter (underside, outboard end) and leave (inner face at Starlink / PV1-2).
- Tent (R27/R28): T1A (230 V) and T1D (12 V + PV) in separate spiral wraps, saddled under the tent base and down the lid front.
- T2 crosses from the tent top to the base at the FRONT HINGE with a 150 mm service loop: the shell opens about that hinge.

12 V + / DC · DC – · 230 V (sheathed flex) · PV · data / control · earth · tent harness (detachable) · R07 run code (schedule on E8) · P-clip / saddle

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E13 of E16**
TITLE **Cable routes: cab–tub gap, junction box, roof and tent harnesses**	REV / DATE **v0.14 · 2026-10-23**

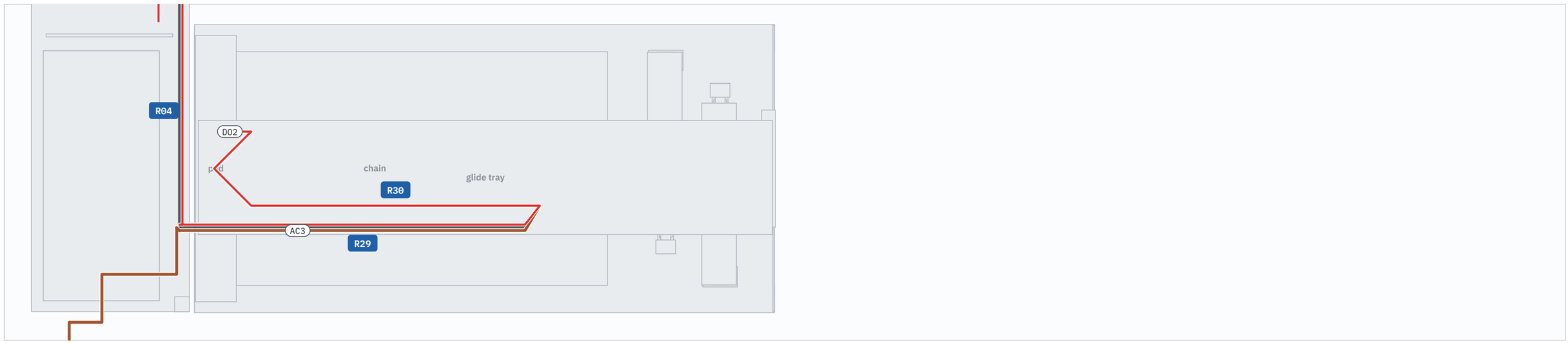
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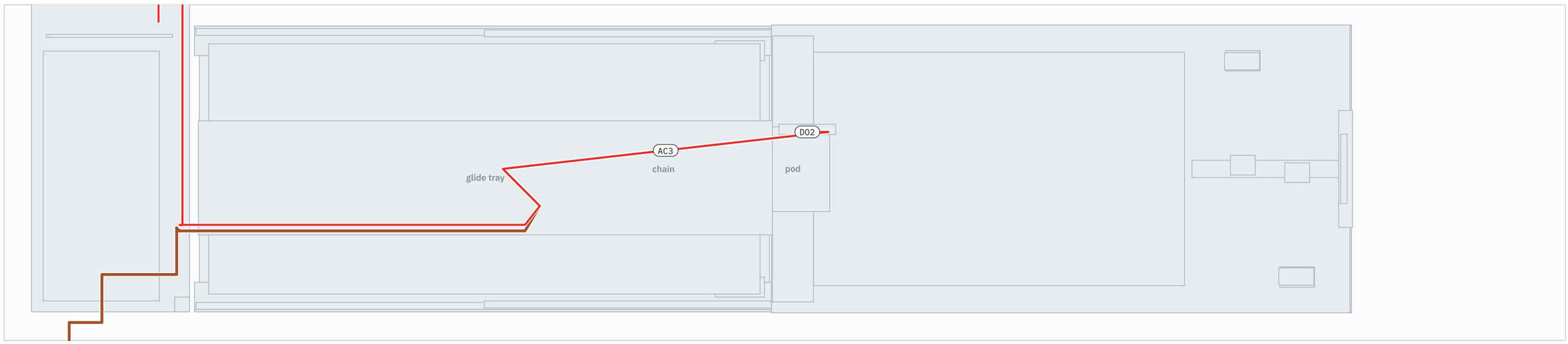
Cable routes: power slide (trunk and chain)

AC3 (230 V) and DO2 (12 V) leave the bay under the guard, run on the glide tray, then through the chain (separator between them) to the pod.

PLAN — stowed (travel) · 1:8 on A3



PLAN — fully out (camp) · 1:8 on A3



CHAIN

- Chain: igus E2.15 R75 lying on its side on the glide tray; fixed end at mid-travel, moving end 115 mm behind the carriage front; length = travel/2 + πR.
- Cables in the chain: AC3 2.5 mm² 3-core high-flex (chain rated) and DO2 6 mm² twin high-flex, with an igus separator between them.
- Strain relief (tie plates) at both chain ends; 600 mm of each cable is in the chain (included in the schedule lengths).
- Inspect with the slide fully out: tray, fixed run, loop and both brackets are open from above.

12 V + / DC · DC − · 230 V (sheathed flex) · PV · data / control · earth · tent harness (detachable) · R07 run code (schedule on E8) · P-clip / saddle

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E14 of E16**
TITLE **Cable routes: power slide (trunk and chain)**	REV / DATE **v0.14 · 2026-10-23**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Bundle cross-sections and fill

Every run's cross-section as packed in the model (biggest cables first; twins as two cores). Dashed circle = conduit inside diameter.

R01 Bay riser, busbar side



11 cables · 723 mm²

AC1 DCF AC5 CHB SW1 D1 SW2 DO2 MPV HTR FAN

R02 Bay rear channel, high



9 cables · 695 mm²

AC1 DCF AC3 AC5 CHB SW1 D1 SW2 FAN

R03 Bay rear channel, cross link



1 cables · 46 mm²

MPV

R04 Bay rear channel, low (to the slid



2 cables · 132 mm²

AC3 DO2

R05 Bay exit over the inverter



17 cables · 1162 mm²

AC1 DCF AC3 AC5 ALT CHB D2 PVR SW1 PVS PVT D1 SW2 MPR ORO MPT

R31 Bay front channel (chargers to the



3 cables · 278 mm²

R06 Charging-board loom



9 cables · 632 mm²

ALT MPR ORO D2 PVR PVT D4 IGN VIG

R07 Tray, 230 V compartment



3 cables · 286 mm²

AC1 AC3 AC5

R08 Tray, 12 V compartment



14 cables · 875 mm²

DCF ALT CHB D2 PVR SW1 PVS PVT D1 SW2 D4 D5 IGN V…AC1 AC3 AC4 AC5

R09 Wall channel, 230 V drop



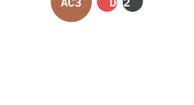
4 cables · 373 mm²

R10 Wall channel, 12 V drop

14 cables · 875 mm²

DCF ALT CHB D2 PVR SW1 PVS PVT D1 SW2 D4 D5 IGN V…ALT CHB PVR DO5 VIS VSW PVS PVT DO3 PE2 DO4 IGN V…AC1 AC3 AC4

R32 Gland plate, 12 V rows



13 cables · 785 mm²

ALT CHB PVR DO5 VIS VSW PVS PVT DO3 PE2 DO4 IGN V…AC1 AC3 AC4

R11 Cabinet, 230 V entry

3 cables · 286 mm²

R12 Cabinet entry (12 V slot)

13 cables · 689 mm²

DCF D2 DO5 VIS SW1 VSW D1 SW2 DO3 PE2 DO4 D4 D5

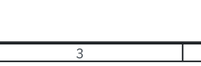
R33 Cabinet top loom



15 cables · 799 mm²

DCF D3 D2 DO5 VIS D1 D01A D01B D03 D01C D01D D04 …D3 SW1 VSW SW2 D01A D01B D01C D01D

R13 Rear-panel loom



8 cables · 319 mm²

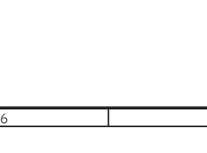
R14 Wall cavity + under floor, 230 V



2 cables · 173 mm² · fill 35% of 25 mm

AC4 AC5

R15 Wall cavity + under floor, 12 V



13 cables · 785 mm² · fill 40% of 50 mm

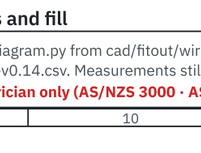
ALT CHB PVR DO5 VIS VSW PVS PVT DO3 PE2 DO4 IGN V…AC4 AC5

R16 Chassis rail, 230 V, tub to the ga



2 cables · 173 mm² · fill 35% of 25 mm

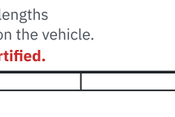
R17 Chassis rail, 12 V, tub to the gap



11 cables · 677 mm² · fill 34% of 50 mm

ALT PVR DO5 VIS VSW PVS PVT DO3 DO4 IGN VIG

R18 Chassis rail, 12 V, gap to the cab



3 cables · 221 mm² · fill 45% of 25 mm

ALT DO3 IGN

R19 Chassis rail, 12 V, cabin to the e



2 cables · 176 mm² · fill 36% of 25 mm

ALT IGN

R20 Cabin branch



1 cables · 46 mm² · fill 35% of 13 mm

DO3

R21 Engine-bay riser



2 cables · 176 mm² · fill 36% of 25 mm

ALT IGN

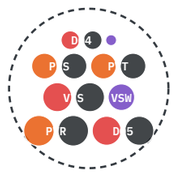
R22 Gap riser, 230 V



2 cables · 173 mm² · fill 35% of 25 mm

AC4 AC5

R23 Gap riser, 12 V



8 cables · 455 mm² · fill 36% of 40 mm

PVR DO5 VIS VSW PVS PVT DO4 VIG

R24 Junction box pass-through



6 cables · 310 mm²

PVR VIS VSW PVS DO4 VIG

R25 Cab back riser to the roof



6 cables · 310 mm² · fill 39% of 32 mm

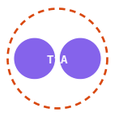
PVR VIS VSW PVS DO4 VIG

R26 Platform rear end member

2 cables · 124 mm²

PVR DO4

R27 Tent harness, 230 V



1 cables · 173 mm² · fill 35% of 25 mm

T1A

R28 Tent harness, 12 V + PV

1 cables · 214 mm² · fill 44% of 25 mm

T1D

R29 Slide trunk (fixed, on the glide t



2 cables · 132 mm²

AC3 DO2

R30 Power-slide cable chain

2 cables · 132 mm²

AC3 DO2

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E15 of E16**
TITLE **Bundle cross-sections and fill**	REV / DATE **v0.14 · 2026-10-23**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

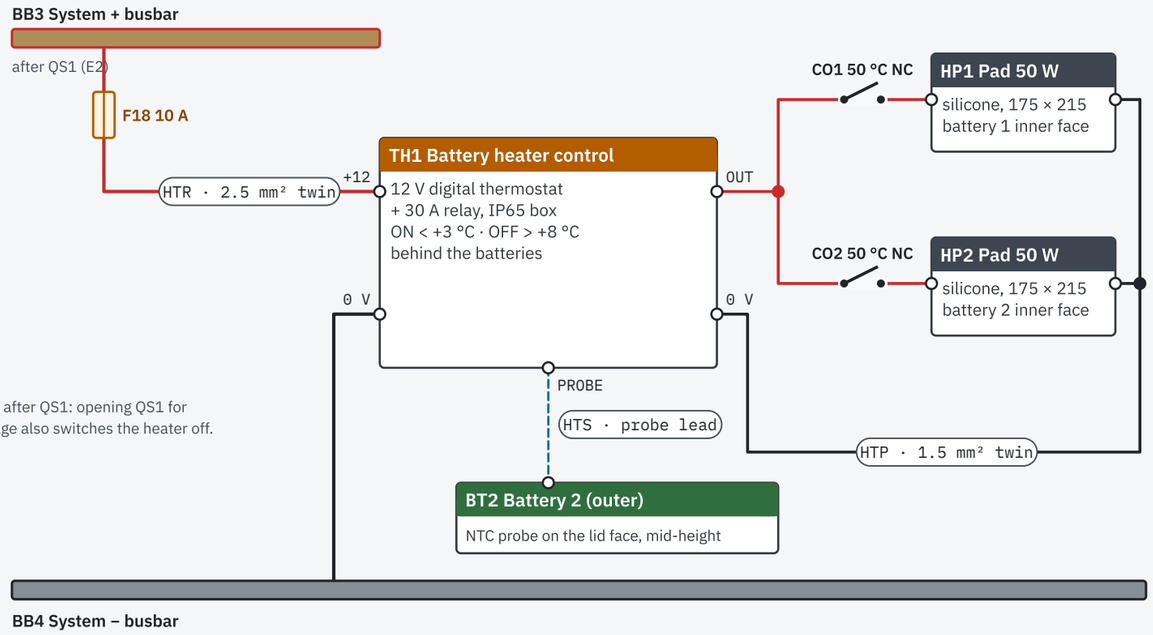
230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Cold-weather heating: battery bank and water can

Snow camping (−5 to −15 °C nights): keep the LiFePO₄ bank above 0 °C so it may charge, stop every charger below +3 °C, and keep the 30 L can from freezing.

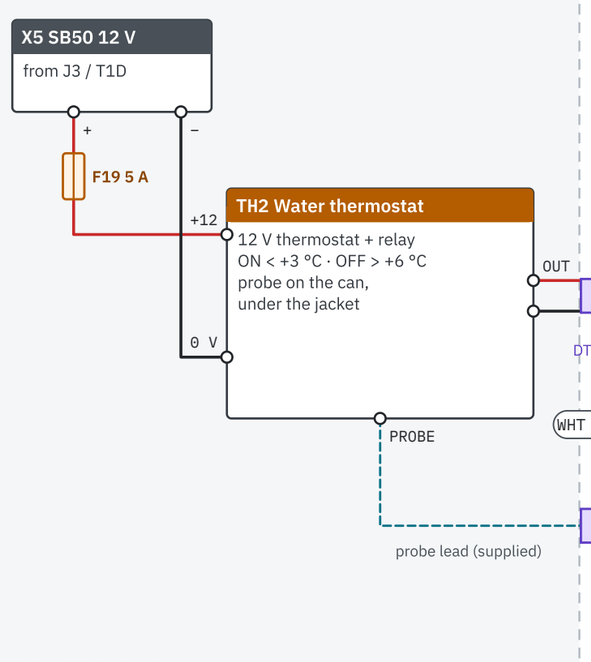
FRONT BAY: BATTERY HEATER

foam on 5 faces; 2 × 50 W pads between the batteries



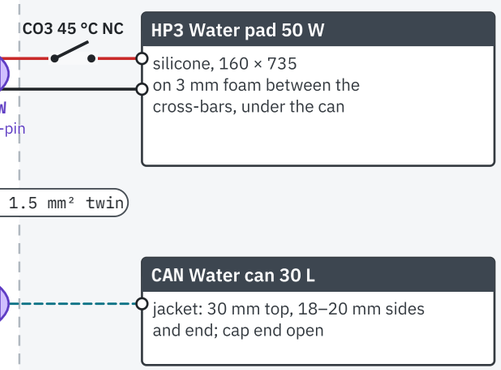
TENT SERVICES BOX

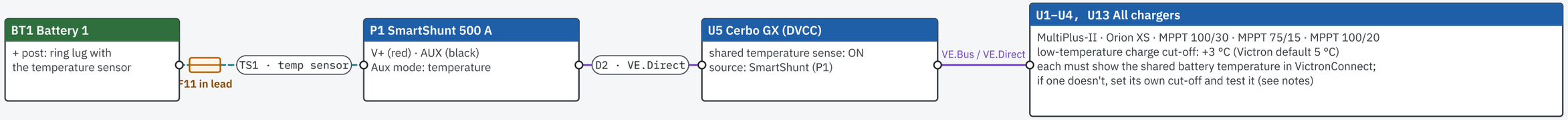
stays with the tent · 12 V from J3 (T1D, E7)



LOAD LID, UNDER THE TENT

vehicle side





CHARGE PROTECTION: NO CHARGING BELOW +3 °C

backstop if a heater fails or the bank is cold at start-up

SETTINGS

- TH1: ON below +3 °C, OFF above +8 °C (probe on BT2, the outer, colder battery).
- TH2: ON below +3 °C, OFF above +6 °C.
- Victron: DVCC on, shared temperature sense from P1, every charger's low-temp cut-off +3 °C.
- Ask iTechworld whether the BMS blocks charging below 0 °C; the build doesn't rely on it.

ENERGY PER NIGHT [EST]

- Bank (insulated, 4 W/K): holding +3 °C uses ~23 Ah on a −5 °C night, ~43–60 Ah at −10 to −15.
- Water can (jacketed, ~2.5 W/K): ~5 Ah at −5 °C, ~15–20 Ah on a hard night.
- 2 × 50 W warms a cold bank −10 → +5 °C in ~3–4 h.
- Daily winter use is ~210 Ah of 800 Ah: plan driving or shore power every 2–3 days.

SAFETY

- Each pad has a normally-closed cut-out bonded to it, in series: a stuck relay can't overheat a battery case (50 °C) or the can (45 °C).
- Pads only on the batteries' sides, between the cage straps; never under a battery.
- 0.13 W/cm² on the case: confirm with iTechworld that the case is fine at 50 °C locally.

TEST ONCE, BEFORE WINTER

1. Chill the TS1 lug (ice pack in a bag) below +3 °C: every charger must stop or show 'low temperature' (forum reports are mixed).
2. Chill each probe: the relay must click on; each pad draws ~4.2 A (clamp meter).
3. Warm each pad's cut-out with a heat gun: the pad must switch off.

LEGEND · line weight ∝ conductor size · ID · size tags = cable schedule IDs · purple ▶◀ = detachable connector · yellow flags = continues on another sheet

DC + (red) · DC − (black) · Data (VE.Bus / VE.Direct / USB) · Signal / sense

PROJECT **Kia Tasman X-Pro fit-out · electrical**	SHEET **E16 of E16**
TITLE **Cold-weather heating: battery bank and water can**	REV / DATE **v0.14 · 2026-10-23**

Generated by tools/wiring/diagram.py from cad/fitout/wiring.py — cable IDs and lengths match bom/cable-schedule-v0.14.csv. Measurements still [EST] until confirmed on the vehicle.

230 V work: licensed electrician only (AS/NZS 3000 · AS/NZS 3001.2). Not certified.

Schedules

Cable schedule

Lengths are measured along each cable's routed path in the 3D model, plus 10 % slack and 0.3 m for terminations.

ID	CIRCUIT	FROM	TO	CABLE	PROTECTION	M	ROUTE (RUNS, IN ORDER)	HARNESS
B1P	Bank +	Battery 1 + (MRBF 300 A)	Bank + busbar	70 mm² tinned, red	MRBF 300 A on terminal	0.85	local	vehicle
B1N	Bank -	Battery 1 -	Bank - busbar	70 mm² tinned, black	-	1.35	local	vehicle
B2P	Bank +	Battery 2 + (MRBF 300 A)	Bank + busbar	70 mm² tinned, red	MRBF 300 A on terminal	0.85	local	vehicle
B2N	Bank -	Battery 2 -	Bank - busbar	70 mm² tinned, black	-	1.35	local	vehicle
RB1	Main +	Bank + busbar	Remote battery switch IN	120 mm² tinned, red	2 x MRBF 300 A upstream (parallel)	0.58	local	vehicle
RB2	Main +	Remote battery switch OUT	System + busbar	120 mm² tinned, red	-	0.65	local	vehicle
SH1	Main -	Bank - busbar	SmartShunt (battery side)	120 mm² tinned, black	-	0.42	local	vehicle
SH2	Main -	SmartShunt (system side)	System - busbar	120 mm² tinned, black	-	0.59	local	vehicle
CT1	Inverter +	System + busbar	Class-T 400 A	2 x 50 mm² tinned, red	-	1.06	local	vehicle
INVP	Inverter +	Class-T 400 A	MultiPlus-II DC +	2 x 50 mm² tinned, red	Class-T 400 A	1.44	local	vehicle
INVN	Inverter -	System - busbar	MultiPlus-II DC -	2 x 50 mm² tinned, black	-	0.98	local	vehicle
ORO	Charging output	Orion XS 12/12-50A	System +/- busbars (MIDI fuse)	16 mm² twin	MIDI 70 A	1.46	R06 > R31	vehicle
MPR	Charging output	MPPT 100/30 (roof)	System +/- busbars (MIDI fuse)	16 mm² twin	MIDI 40 A	1.25	R06 > R31	vehicle
MPT	Charging output	MPPT 75/15 (tent)	System +/- busbars (MIDI fuse)	6 mm² twin	MIDI 40 A (shared with MPV)	1.02	R31	vehicle
MPV	Charging output	MPPT 100/20 (visor)	System +/- busbars (MIDI fuse)	6 mm² twin	MIDI 40 A (shared with MPT)	1.15	R03 > R01	vehicle
DCF	DC distribution	System + busbar (MEGA 100 A)	Rear DC fuse block	35 mm² twin (red/black)	MEGA 100 A	4.1	R01 > R02 > R05 > R08 > R10 > R12 > R33	vehicle
AC1	AC out	MultiPlus-II AC-out 1	RCBO enclosure (Q0 main switch)	4 mm² 3-core flex (TPS-flex)	MultiPlus internal limit + 15 A AC-in limit; Q0 2-pole main switch	3.97	R01 > R02 > R05 > R07 > R09 > R11	vehicle

ID	CIRCUIT	FROM	TO	CABLE	PROTECTION	M	ROUTE (RUNS, IN ORDER)	HARNESS
AC2	Rear GPO	RCBO Q1	Rear panel double GPO	2.5 mm² 3-core flex	RCBO 10 A 30 mA	0.45	local	vehicle
AC3	Slide GPO	RCBO Q2	Power-slide pod GPO	2.5 mm² 3-core high-flex (chain rated)	RCBO 10 A 30 mA	5.76	R11 > R09 > R07 > R05 > R02 > R04 > R29 > R30	vehicle
AC4	Tent heater GPO	RCBO Q3	Gap junction J1 -> tent services box outlet	2.5 mm² 3-core flex, UV rated	RCBO 10 A 30 mA	4.45	R11 > R09 > R14 > R16 > R22	vehicle
AC5	Shore inlet	Gap junction J2 (from the tent's 15 A inlet)	MultiPlus-II AC-in	2.5 mm² 3-core flex (15 A)	RCBO 16 A 30 mA 2-pole in the services box + MultiPlus AC-in limit 15 A	7.36	R22 > R16 > R14 > R09 > R07 > R05 > R02 > R01	vehicle
PE2	AC earth to chassis	RCBO enclosure earth bar	Chassis bond point (passenger rail, near the gland)	6 mm² green/yellow	-	1.91	R12 > R32 > R15	vehicle
DO1A	Rear Anderson 1	DC fuse block	Rear panel Anderson 1	6 mm² twin	ATC 30 A	1.03	R33 > R13	vehicle
DO1B	Rear Anderson 2	DC fuse block	Rear panel Anderson 2	6 mm² twin	ATC 30 A	1.09	R33 > R13	vehicle
DO1C	Rear USB-C x2	DC fuse block	Rear panel USB-C x2	4 mm² twin	ATC 25 A	0.96	R33 > R13	vehicle
DO1D	Rear 12 V socket	DC fuse block	Rear panel 12 V socket	4 mm² twin	ATC 15 A	1.05	R33 > R13	vehicle
DO1E	Rear USB-C link	USB-C U7 terminals	USB-C U8 terminals	4 mm² twin	ATC 25 A (DO1C)	0.42	local	vehicle
DO2	Slide DC	System + busbar spare stud (MRBF 30 A); - to the system - busbar	Power-slide pod (FB2)	6 mm² twin high-flex	MRBF 30 A (F17); FB2: X3 20 A, X7 15 A	3.57	R01 > R04 > R29 > R30	vehicle
DO3	Cabin compute feed	DC fuse block	Compute box SB50 inlet (under the rear seat)	6 mm² twin	ATC 15 A (+ BatteryProtect and per-device fuses in the box)	4.79	R33 > R12 > R32 > R15 > R17 > R18 > R20	vehicle
DO4	Starlink Mini	DC fuse block	Starlink Mini (roof)	4 mm² twin + DC barrel lead	ATC 7.5 A	5.77	R33 > R12 > R32 > R15 > R17 > R23 > R24 > R25 > R26	vehicle
DO5	Tent 12 V	DC fuse block	Gap junction J3 -> tent (SB50 red)	10 mm² twin	ATC 20 A	4.46	R33 > R12 > R32 > R15 > R17 > R23	vehicle
ALT	Alternator charge	Start battery + (F8 MIDI 60 A); - to the body earth	Orion XS input	25 mm² twin (+/-), tinned	MIDI 60 A at the start battery	9.05	R21 > R19 > R18 > R17 > R15 > R32 > R10 > R08 > R05 > R06	vehicle
IGN	Orion ignition sense	Engine-bay fuse box (add-a-fuse, ignition)	Orion XS remote H	1.0 mm² single	ATC 2 A add-a-fuse	9.37	R21 > R19 > R18 > R17 > R15 > R32 > R10 > R08 > R05 > R06	vehicle
PVR	Roof PV	Roof panels PV1 + PV2 (parallel, MC4 Y-branch)	QP1 -> MPPT 100/30 PV in	10 mm² solar pair (UV)	2-pole DC PV isolator QP1 32 A	7.98	R26 > R25 > R24 > R23 > R17 > R15 > R32 > R10 > R08 > R05 > R06	vehicle
PVT	Tent PV	Gap junction J4 (from the tent panels)	QP2 -> MPPT 75/15 PV in	6 mm² solar pair (UV)	2-pole DC PV isolator QP2 32 A	6.46	R23 > R17 > R15 > R32 > R10 > R08 > R05 > R06	vehicle

ID	CIRCUIT	FROM	TO	CABLE	PROTECTION	M	ROUTE (RUNS, IN ORDER)	HARNESS
PVS	Visor PV	U11 box (MC4 couplers at the visor chain end)	QP3 -> MPPT 100/20 (visor) PV in	6 mm² solar pair (UV)	2-pole DC PV isolator QP3 32 A	7.25	R25 > R24 > R23 > R17 > R15 > R32 > R10 > R08 > R05	vehicle
SW1	Battery switch control	Rear panel ML-RBS switch (Blue Sea 2155)	ML-RBS control harness	4-core 1.0 mm² (+12 V feed, control, LED, ground)	2 A inline at bank + (RBF)	3.75	R13 > R12 > R10 > R08 > R05 > R02 > R01	vehicle
RBF	Battery switch control	Bank + busbar	ML-RBS control feed (inline 2 A)	1.0 mm² red	ATC 2 A inline	0.57	local	vehicle
TS1	Shunt sense + battery temperature	Battery 1 + post (lug with sensor, under the MRBF)	SmartShunt Vbatt+ and Aux	Victron temperature sensor lead (2-core, supplied)	fuse in the lead (supplied)	0.89	local	vehicle
HTR	Battery heater feed	System + busbar (F18 10 A inline); - to the system - busbar	Heater controller TH1	2.5 mm² twin	ATC 10 A inline (F18)	0.93	R01	vehicle
HTP	Battery heater pads	Heater controller TH1 (relay out)	Heater pads HP1 + HP2 (between the batteries)	1.5 mm² twin (silicone)	F18 (upstream); 50 C cut-out on each pad	0.44	local	vehicle
HTS	Battery heater probe	Heater controller TH1 (sensor in)	NTC probe on battery 2 (lid face, mid-height)	NTC probe lead (supplied, 2-core)	-	0.66	local	vehicle
FAN	Inverter fan	System + busbar (inline 5 A)	92 mm thermostatic fan	1.5 mm² twin	ATC 5 A inline	1.35	R01 > R02	vehicle
CGX	Cerbo GX power	DC fuse block	Cerbo GX power in	0.75 mm² twin	ATC 2 A	0.74	R33	vehicle
PE1	Inverter case earth	MultiPlus-II chassis earth stud	System - busbar	50 mm² green/yellow	-	0.68	local	vehicle
CHB	Chassis bond	System - busbar	Chassis bond point (passenger rail, near the gland)	25 mm² black	-	4.7	R01 > R02 > R05 > R08 > R10 > R32 > R15	vehicle
SW2	Inverter remote on/off	Rear panel switch S2	MultiPlus-II remote terminals	2-core 0.75 mm²	-	3.99	R13 > R12 > R10 > R08 > R05 > R02 > R01	vehicle
D1	VE.Bus	Cerbo GX	MultiPlus-II	RJ45 VE.Bus cable	-	4.36	R33 > R12 > R10 > R08 > R05 > R02 > R01	vehicle
D2	VE.Direct	Cerbo GX VE.Direct 1-3	SmartShunt / MPPT 100/30 / MPPT 75/15	VE.Direct cables x3	-	3.55	R33 > R12 > R10 > R08 > R05 > R06	vehicle
D4	VE.Direct	Cerbo GX USB 1	Orion XS VE.Direct	VE.Direct to USB interface	-	3.29	R33 > R12 > R10 > R08 > R05 > R06	vehicle
D5	VE.Direct	Cerbo GX USB 2	MPPT 100/20 (visor) VE.Direct	VE.Direct to USB interface	-	3.56	R33 > R12 > R10 > R08 > R05	vehicle
D3	Display	Cerbo GX (HDMI + USB)	GX Touch 50	GX Touch cable (HDMI + USB)	-	1.01	R33 > R13	vehicle
PVV	Visor PV (moving)	Visor panel junction box	U11 box (MC4 couplers)	4 mm² solar pair, high-flex (chain rated)	-	2.29	local	vehicle

ID	CIRCUIT	FROM	TO	CABLE	PROTECTION	M	ROUTE (RUNS, IN ORDER)	HARNESS
VIS	Visor drive power	DC fuse block (way 10)	Visor controller U11 (roof)	10 mm² twin	ATC 20 A	5.3	R33 > R12 > R32 > R15 > R17 > R23 > R24 > R25	vehicle
VSW	Visor control	Rear panel visor button (S3)	Visor controller U11 (roof)	4-core 0.75 mm² (deploy, stow, LED, common)	-	5.17	R13 > R12 > R32 > R15 > R17 > R23 > R24 > R25	vehicle
VIG	Visor interlock	Orion XS remote H (ignition, F9)	Visor controller interlock input	1.0 mm² single	F9 2 A (shared)	6.93	R06 > R05 > R08 > R10 > R32 > R15 > R17 > R23 > R24 > R25	vehicle
T1A	Tent harness 230 V	Tent services box (heater outlet + shore inlet)	Gap junction J1 + J2	2 x 2.5 mm² 3-core flex, sheathed together, UV rated	Q4 2-pole RCBO 16 A (AC-in); Q3 (AC-out)	3.82	R27	tent
T1D	Tent harness 12 V + PV	Tent services box (SB50 + PV bulkhead)	Gap junction J3 + J4	10 mm² twin (12 V) + 6 mm² solar pair (PV), sheathed together	DO5 20 A (12 V)	1.92	R28	tent
T2	Tent PV harness	Tent-top junction box (PV3 + PV4 in series)	Services box PV bulkhead	6 mm² solar pair (UV)	-	1.68	local	tent
WHT	Water can heater	Tent services box (F19 5 A + thermostat TH2)	Water can pad HP3 (under the can)	1.5 mm² twin (silicone), plug-in	ATC 5 A (F19) in the services box; 45 C cut-out on the pad	0.54	local	tent

Cable to buy

CABLE	MODEL M	BUY M (+15 %)
0.75 mm² twin	0.7	1
1.0 mm² red	0.6	1
1.0 mm² single	16.3	19
1.5 mm² twin	1.4	2
1.5 mm² twin (silicone)	0.4	1
1.5 mm² twin (silicone), plug-in	0.5	1
10 mm² solar pair (UV)	8.0	10
10 mm² twin	9.8	12
10 mm² twin (12 V) + 6 mm² solar pair (PV), sheathed together	1.9	3
120 mm² tinned, black	1.0	2
120 mm² tinned, red	1.2	2
16 mm² twin	2.7	4
2 x 2.5 mm² 3-core flex, sheathed together, UV rated	3.8	5
2 x 50 mm² tinned, black	1.0	2
2 x 50 mm² tinned, red	2.5	3
2-core 0.75 mm²	4.0	5

CABLE	MODEL M	BUY M (+15 %)
2.5 mm² 3-core flex	0.5	1
2.5 mm² 3-core flex (15 A)	7.4	9
2.5 mm² 3-core flex, UV rated	4.5	6
2.5 mm² 3-core high-flex (chain rated)	5.8	7
2.5 mm² twin	0.9	2
25 mm² black	4.7	6
25 mm² twin (+/-), tinned	9.1	11
35 mm² twin (red/black)	4.1	5
4 mm² 3-core flex (TPS-flex)	4.0	5
4 mm² solar pair, high-flex (chain rated)	2.3	3
4 mm² twin	2.4	3
4 mm² twin + DC barrel lead	5.8	7
4-core 0.75 mm² (deploy, stow, LED, common)	5.2	6
4-core 1.0 mm² (+12 V feed, control, LED, ground)	3.8	5
50 mm² green/yellow	0.7	1
6 mm² green/yellow	1.9	3
6 mm² solar pair (UV)	15.4	18
6 mm² twin	9.1	11
6 mm² twin high-flex	3.6	5
70 mm² tinned, black	2.7	4
70 mm² tinned, red	1.7	2
GX Touch cable (HDMI + USB)	1.0	2
NTC probe lead (supplied, 2-core)	0.7	1
RJ45 VE.Bus cable	4.4	6
VE.Direct cables x3	3.5	5
VE.Direct to USB interface	6.8	8
Victron temperature sensor lead (2-core, supplied)	0.9	2

Route schedule

Each physical run: where it goes, how the cables are held, and which cables share it. Fill is the cables' cross-section over the conduit's (≤ 50 % for split conduit that the cables are laid into).

RUN	NAME	M	N	CONTAINMENT	FILL	V	FIXING	CABLES
R01	Bay riser, busbar side	0.38	11	open		12 V	rear channel behind the busbars (gap to battery 1); cable ties to the rear posts' saddle bar every 150 mm	MPV DCF AC1 AC5 DO2 SW1 HTR FAN CHB SW2 D1
R02	Bay rear channel, high	0.58	9	open		12 V	behind the distribution board edge and the inverter, under the top rail; adhesive tie mounts on the guard frame	DCF AC1 AC3 AC5 SW1 FAN CHB SW2 D1
R03	Bay rear channel, cross link	0.11	1	open		12 V	behind the board edge, visor MPPT output	MPV
R04	Bay rear channel, low (to the slide)	0.85	2	open		12 V	along the base tray behind the inverter, then down from the high channel; P-clips to the base tray	AC3 DO2
R05	Bay exit over the inverter	0.33	17	open		12 V	across the top of the inverter compartment and over the passenger drawer carcass; grommet strip on the carcass edge	DCF AC1 AC3 AC5 ALT IGN PVR PVT PVS SW1 CHB SW2 D1 D2 D4 D5 VIG
R31	Bay front channel (chargers to the MIDI fuses)	0.36	3	open		12 V	between the front wall and the distribution board's front edge, under the charging board; tie mounts on the board edge	ORO MPR MPT
R06	Charging-board loom	0.67	9	open		12 V	down the front of the inverter compartment and along under the chargers' terminal faces; tie mounts on the board	ORO MPR ALT IGN PVR PVT D2 D4 VIG
R07	Tray, 230 V compartment	1.42	3	tray		230 V	aluminium tray, 230 V side of the divider; ties through the tray slots every 120 mm	AC1 AC3 AC5
R08	Tray, 12 V compartment	1.26	14	tray		12 V	aluminium tray, 12 V side of the divider; ties through the tray slots every 120 mm	DCF ALT IGN PVR PVT PVS SW1 CHB SW2 D1 D2 D4 D5 VIG
R09	Wall channel, 230 V drop	0.2	4	channel		230 V	22 mm channel behind the drawer runner panel, then out through the 230 V grommet row of the gland plate; ties to adhesive mounts	AC1 AC3 AC4 AC5
R10	Wall channel, 12 V drop	0.18	14	channel		12 V	22 mm channel behind the drawer runner panel (two layers); ties to adhesive mounts on the tub wall	DCF ALT IGN PVR PVT PVS SW1 CHB SW2 D1 D2 D4 D5 VIG
R32	Gland plate, 12 V rows	0.02	13	open		12 V	12 V grommet rows of the gland plate in the old outlet hole	PE2 DO3 DO4 DO5 ALT IGN PVR PVT PVS CHB VIS VSW VIG
R11	Cabinet, 230 V entry	0.22	3	open		230 V	LV gland in the cabinet top, down the outboard corridor to the RCBO side gland	AC1 AC3 AC4
R12	Cabinet entry (12 V slot)	0.22	13	open		12 V	down the wall channel and through the 12 V slot (split grommet) in the drawer floor and cabinet top	DCF PE2 DO3 DO4 DO5 SW1 SW2 D1 D2 D4 D5 VIS VSW
R33	Cabinet top loom	0.36	15	open		12 V	forward along the outboard corridor beside the RCBO enclosure, under the cabinet top; tie mounts	DCF DO1A DO1B DO1C DO1D DO3 DO4 DO5 CGX D1 D2 D4 D5 D3 VIS
R13	Rear-panel loom	0.26	8	open		12 V	vertical loom between the RCBO enclosure lid and the rear-panel devices; spiral wrap, ties to the door stile	DO1A DO1B DO1C DO1D SW1 SW2 D3 VSW
R14	Wall cavity + under floor, 230 V	1.09	2	orange corrugated conduit 25 mm ID	35%	230 V	LV grommet row in the gland plate; orange corrugated conduit down the wall cavity and under the floor to the rail	AC4 AC5
R15	Wall cavity + under floor, 12 V	1.02	13	split conduit 50 mm ID	40%	12 V	12 V grommet row in the gland plate; black split corrugated conduit down the wall cavity and under the floor to the rail	PE2 DO3 DO4 DO5 ALT IGN PVR PVT PVS CHB VIS VSW VIG
R16	Chassis rail, 230 V, tub to the gap	1.37	2	orange corrugated conduit 25 mm ID	35%	230 V	orange corrugated conduit on the rail web, below the 12 V conduit; rubber-lined P-clips every 300 mm	AC4 AC5

RUN	NAME	M	N	CONTAINMENT	FILL	V	FIXING	CABLES
R17	Chassis rail, 12 V, tub to the gap	1.43	11	split conduit 50 mm ID	34%	12 V	black split conduit on the rail web; rubber-lined P-clips into M6 rivnuts every 300 mm (no welding or drilling the flanges)	DO3 DO4 DO5 ALT IGN PVR PVT PVS VIS VSW VIG
R18	Chassis rail, 12 V, gap to the cabin	0.68	3	split conduit 25 mm ID	45%	12 V	black split conduit; P-clips every 300 mm	DO3 ALT IGN
R19	Chassis rail, 12 V, cabin to the engine bay	2.12	2	split conduit 25 mm ID	36%	12 V	black split conduit (heat-rated near the exhaust); P-clips every 300 mm	ALT IGN
R20	Cabin branch	0.25	1	split conduit 13 mm ID	35%	12 V	up to a sealed grommet in the floor under the rear seat (sealant + rubber grommet), then to the compute box SB50 inlet	DO3
R21	Engine-bay riser	0.51	2	split conduit 25 mm ID	36%	12 V	heat-rated split conduit up the inner guard behind the start battery; P-clips to existing guard holes	ALT IGN
R22	Gap riser, 230 V	0.84	2	orange corrugated conduit 25 mm ID	35%	230 V	orange conduit up the cab-tub gap to the LV gland in the junction box base	AC4 AC5
R23	Gap riser, 12 V	0.67	8	split conduit 40 mm ID	36%	12 V	black conduit up the cab-tub gap to the ELV gland in the junction box base	DO4 DO5 PVR PVT PVS VIS VSW VIG
R24	Junction box pass-through	0.18	6	open		12 V	through the junction box on the 12 V side of its barrier (roof cables are not detachable)	DO4 PVR PVS VIS VSW VIG
R25	Cab back riser to the roof	0.57	6	split conduit 32 mm ID	39%	12 V	black conduit up the back of the cab (C-pillar corner), stick-on cable clamps every 250 mm, over the roof edge under the platform	DO4 PVR PVS VIS VSW VIG
R26	Platform rear end member	0.72	2	hollow		12 V	inside the hollow rear end member (34 x 34 mm), in through a grommeted hole in its underside, out through grommets on its inner face	DO4 PVR
R27	Tent harness, 230 V	1.45	1	spiral wrap 25 mm	35%	230 V	orange sheathed harness in UV-stable spiral wrap, saddles under the tent base and on the lid front	T1A
R28	Tent harness, 12 V + PV	1.46	1	spiral wrap 25 mm	44%	12 V	black sheathed harness in spiral wrap, saddles under the tent base and on the lid front	T1D
R29	Slide trunk (fixed, on the glide tray)	0.71	2	open		12 V	under the bay guard into the front of the cassette, along the glide tray outboard of the fixed run; ties to the tray lip	AC3 DO2
R30	Power-slide cable chain	0.84	2	chain		12 V	igus E2.15 (R75) lying on its side, with a separator between the 230 V and 12 V cables; strain-relief at both ends	AC3 DO2

Protection

REF	DEVICE	RATING	LOCATION	PROTECTS
F1	MRBF terminal fuse (Blue Sea 5191)	300 A	Battery 1 + stud	B1P
F2	MRBF terminal fuse (Blue Sea 5191)	300 A	Battery 2 + stud	B2P
F3	Class-T fuse + holder (Blue Sea 5502)	400 A	BB3 system + bus	CT1, INVP (MultiPlus DC)
F4	MIDI fuse	70 A	BB3 stud	ORO (Orion XS output)

FB1 fuse block ways

WAY	CIRCUIT	ATC	CABLE
1	Rear Anderson 1	30 A	DO1A
2	Rear Anderson 2	30 A	DO1B
3	Rear USB-C ×2	25 A	DO1C
4	Rear 12 V socket	15 A	DO1D
5	spare (pod moved to the bay)	–	–
6	Cabin compute feed	15 A	DO3

REF	DEVICE	RATING	LOCATION	PROTECTS
F5	MIDI fuse	40 A	BB3 stud	MPR (MPPT 100/30 output)
F6	MIDI fuse	40 A	BB3 stud	MPT (MPPT 75/15) and MPV (MPPT 100/20) outputs, 6 mm² each
F7	MEGA fuse	100 A	BB3 stud	DCF (feed to FB1)
F8	MIDI fuse + holder	60 A	Start battery +, engine bay	ALT
F9	Add-a-fuse, ignition circuit	2 A	Engine-bay fuse box	IGN
F10	Inline ATC holder	2 A	BB1 bank + bus	RBF, SW1 (ML-RBS control)
F11	Fuse in the Victron temperature sensor lead (supplied)	–	Battery 1 + post (lug)	TS1 (P1 V+ and AUX)
F12	Inline ATC holder	5 A	BB3 stud	FAN
FB1 1–10	ATC blade fuses in FB1 (Blue Sea 5026)	see E3	Rear control cabinet	DO1A–D, DO3–DO5, CGX, VIS (not DO2)
F13	ATC in FB2	15 A	Power slide pod	pod 12 V socket X7
F14	ATC in FB2	20 A	Power slide pod	pod Anderson X3
F17	MRBF terminal fuse (Blue Sea 5191)	30 A	BB3 system + busbar, spare stud	DO2 (power-slide pod)
F18	Inline ATC holder	10 A	BB3 stud 4, front bay	HTR, HTP (battery heater, 2 × 50 W)
F19	Inline ATC holder	5 A	Tent services box (12 V from X5)	WHT (water-can pad, 50 W)
C01–C03	Thermal cut-out, normally closed, bonded to each pad	50 / 50 / 45 °C	On HP1, HP2, HP3	pad over-temperature
FB3	ATC fuse block behind a Smart BatteryProtect	per device	Compute box, under the rear seat	HA PC, router, PoE switch, fan
Q0	Main switch, 2-pole isolator (AS/NZS 3001.2)	32 A	RCBO enclosure, rear cabinet	inverter supply AC1
Q1–Q3	RCBO 1P+N, 30 mA	10 A	RCBO enclosure, rear cabinet	AC2, AC3, AC4
Q4	RCBO 2-pole, 30 mA	16 A	Tent services box	shore inlet → AC-in (T1A, AC5)
QS1	ML-RBS remote battery switch (Blue Sea 7713)	500 A	Front bay	isolation, bank + → system +

WAY	CIRCUIT	ATC	CABLE
7	Starlink Mini	7.5 A	DO4
8	Tent 12 V	20 A	DO5
9	Cerbo GX	2 A	CGX
10	Visor drive	20 A	VIS
11	spare	–	–
12	spare	–	–

Before power-up

- Measure the real cable runs: these lengths come from [EST] positions in the model.
- Confirm the ML-RBS control-wire colours and switch wiring against the Blue Sea 7713 instructions.
- Set every charger to iTechworld's $LiFePO_4$ voltages; set the Orion XS input limit to 25–30 A.
- Torque every stud, then re-check after the first corrugated road.
- Electrician: earth continuity, polarity, RCD trip tests on shore power and in inverter mode.

REF	DEVICE	RATING	LOCATION	PROTECTS
QP1–QP3	2-pole DC PV isolator	32 A	Front bay, beside the MPPTs	PVR / PVT / PVS (servicing)
U11 F1–F4	Fuses inside the visor controller	15 / 7.5 / 7.5 / 3 A	U11 box, roof	M2, M3, M4, Y1/Y2 (+ 2 A logic)

Component register

REF	ITEM	MODEL	LOCATION	SHEETS
BT1, BT2	House batteries 12.8 V 400 Ah	iTechworld iTech400X Pro	Front bay, on end	E2
BB1–BB4	Busbars (bank ±, system ±)	Blue Sea PowerBar 600 A	Front bay distribution column	E2
QS1	Remote battery switch	Blue Sea ML-RBS 7713 (12 V)	Front bay	E2, E6
P1	Battery monitor	Victron SmartShunt 500 A	Front bay	E2, E6
U1	Inverter/charger	Victron MultiPlus-II 12/3000/120-32 230 V	Front bay, lying flat	E2, E5, E6
U2	DC-DC charger	Victron Orion XS 12/12-50A	Charging board	E4, E6
U3	Solar charger, roof	Victron SmartSolar MPPT 100/30	Charging board	E4, E6
U13	Solar charger, visor	Victron SmartSolar MPPT 100/20	Front bay, passenger face of the distribution board	E4, E6
U4	Solar charger, tent	Victron SmartSolar MPPT 75/15	Charging board	E4, E6
M1	Inverter fan	92 mm 12 V thermostatic	Above the inverter	E2
TH1	Battery heater control	12 V digital thermostat (NTC probe) + 30 A relay, IP65 box	Behind the batteries, on the rear foam	E16
HP1, HP2	Battery heater pads 50 W	12 V silicone pad 175 × 215, self-adhesive, CO1/CO2 50 °C	Facing sides of BT1 / BT2	E16
TT1	Battery temperature sensor	Victron temperature sensor (BMV / SmartShunt), lug on BT1 +	Battery 1 + post	E2, E16
TH2, HP3	Water-can thermostat + pad 50 W	12 V thermostat + relay; silicone pad 160 × 735, CO3 45 °C	Services box / under the can	E16
XW	Water-pad plug	Deutsch DT 4-pin (pad + probe)	Beside the tent services box	E16
FB1	Fuse block 12-way + neg bus	Blue Sea 5026	Rear control cabinet	E3
FB2	Local fuse holder + studs	2-way ATC	Power slide pod	E3
U12, FB3	Compute box: BatteryProtect + 6-way ATC block	Victron SBP-65, Blue Sea 5025-class	Under the rear seat	E3
U5	GX device	Victron Cerbo GX MK2	Rear control cabinet	E3, E6
U6	Display	Victron GX Touch 50	Rear panel	E6
S1	Battery isolate switch	Blue Sea 2155 Contura SPDT with LED	Rear panel	E6
S2	Inverter on/off switch	SPST rocker	Rear panel	E6
S3	Visor deploy / stow button	momentary ON-OFF-ON rocker with LED	Rear panel	E6, E9
RCBO	RCBO enclosure, Q0–Q3	IP66 DIN enclosure, 2-pole main switch + 3 × RCBO 10 A 30 mA	Rear control cabinet	E5
XS1	Rear double GPO	IP66 10 A (Clipsal 56 series)	Rear panel	E5
XS2	Pod GPO	Flush 10 A, faces up	Power slide pod	E5

REF	ITEM	MODEL	LOCATION	SHEETS
XS3	Tent heater outlet	Clipsal 56SO310A IP66 10 A	Tent services box	E5, E7
XI1, Q4	Shore inlet + RCBO	Clipsal 56AI315 15 A + 2-pole RCBO 16 A 30 mA	Tent services box	E5, E7
X1–X5	Anderson SB50 sockets	Anderson SB50 red + flush housing	Rear panel ×2, pod, compute-box inlet, tent box	E3
X6, X7	12 V sockets	Narva / Blue Sea	Rear panel, pod	E3
U7, U8	Dual USB-C sockets, 100 W PD + 20 W each	Treklo dual USB-C 120 W, 12–24 V	Rear panel ×2	E3
JB	Gap junction box	IP66 polycarbonate with an internal barrier (230 V | 12 V)	Cab–tub gap, passenger corner	E7, E13
J1	230 V out connector	IP66 10 A socket (vehicle) / plug (tent)	Gap junction box	E5, E7
J2	230 V in connector	IP66 15 A inlet (vehicle) / extension socket (tent)	Gap junction box	E5, E7
J3	12 V tent connector	Anderson SB50 red	Gap junction box	E3, E7
J4	Tent PV connector	Anderson SB50 grey	Gap junction box	E4, E7
PV1, PV2	Roof panels 190 W + 130 W	Victron BlueSolar (parallel)	Roof platform	E4
PV5	Visor panel ~250 W	semi-flexible on honeycomb backing (own MPPT, U13)	Slide-out visor	E4, E9
U11	Visor controller	12 V PLC / µC + 2 × dual H-bridge, per-channel fuses, IP66 box	Under the platform, rear passenger corner	E3, E9
M2	Slide gearmotor	12 V worm gearmotor, self-locking	Roof platform, rear cross-member	E9
M3, M4	Tilt drives	12 V worm rotary actuators	Hinge beam ends	E9
Y1, Y2	Front latches	12 V solenoid, spring-locked (fail-locked)	Platform front	E9
B1–B4	Limit switches	IP67 roller-lever, NO	Rails and hinge beam	E9
BZ1	Not-stowed buzzer	12 V piezo	Visor controller lid	E9
PV3, PV4	Tent panels 2 × 100 W	Sunman eArc flexible (series)	Tent roof	E4
ST1	Satellite internet	Starlink Mini	Roof platform, inside the frame	E3, E6
BT0, G1, FBX	Vehicle start battery, alternator, fuse box	factory	Engine bay	E4

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